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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
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In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

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The present document is part 5 of a multi-part conformance test specification for Mission Critical services consisting of:

3GPP TS 37.579-1 [2]: "Mission Critical (MC) services; Part 1: Common test environment"

3GPP TS 37.579-2 [3]: "Mission Critical (MC) services; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification"

3GPP TS 37.579-4 [5]: "Mission Critical (MC) services; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification"

3GPP TS 37.579-5: "Mission Critical (MC) services; Part 5: Abstract test suite (ATS)" (the present specification)

3GPP TS 37.579-6 [59]: "Mission Critical (MC) services; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification"

3GPP TS 37.579-7 [60]: "Mission Critical (MC) services; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification"

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the protocol and signalling conformance testing in TTCN-3 for the Mission Critical services defined by 3GPP.

The following TTCN test specification and design considerations can be found in the present document:

- the test system architecture;
- the overall test suite structure;
- the test models and ASP definitions;
- the test methods and usage of communication ports definitions;
- the test configurations;
- the design principles and assumptions;
- TTCN styles and conventions;
- the partial Implementation eXtra Information for Testing (IXIT) proforma;
- the test suites.

The Abstract Test Suites designed in the document are based on the test cases specified in 3GPP TS 37.579-2 [3], TS 37.579-6 [59] and TS 37.579-7 [60].

The applicability of the individual test cases is specified in the test ICS proforma specification in 3GPP TS 37.579-4 [5]). Where appropriate the Abstract Test Suites belonging to the present specification may refer to other Abstract Test Suites e.g. 3GPP TS 36.523-3 [27] and TS 38.523-3 [72] for test requirements related to the 3GPP bearers which carry the Mission Critical services data.

The present document is valid for TTCN development for Mission Critical services clients' conformance tests according to 3GPP Releases starting from Release 14 up to the Release indicated on the cover page of the present document.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document* unless the context in which the reference is made suggests a different Release is relevant (information on the applicable release in a particular context can be found in e.g. test case title, description or applicability, message description or content).

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 37.579-1: "Mission Critical (MC) services; Part 1: Common test environment".

[3] 3GPP TS 37.579-2: "Mission Critical (MC) services; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification".

[4] Void.

[5] 3GPP TS 37.579-4: "Mission Critical (MC) services; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification".

[6] 3GPP TS 36.523-1: "User Equipment (UE) conformance specification; Part 1: Protocol conformance specification"

[7] 3GPP TS 22.179: "Mission Critical Push To Talk (MCPTT) over LTE; Stage 1".

[8] Void.

[9] 3GPP TS 24.379: "Mission Critical Push To Talk (MCPTT) call control; Protocol specification".

[10] 3GPP TS 24.380: "Mission Critical Push To Talk (MCPTT) floor control; Protocol specification".

[11] 3GPP TS 24.481: "Mission Critical Services (MCS) group management; Protocol specification".

[12] 3GPP TS 24.482: "Mission Critical Services (MCS) identity management; Protocol specification".

[13] 3GPP TS 24.483: "Mission Critical Services (MCS) Management Object (MO)".

[14] 3GPP TS 24.484: "Mission Critical Services (MCS) configuration management; Protocol specification".

[15] 3GPP TS 33.179: "Security of Mission Critical Push-To-Talk (MCPTT)".

[16] 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

[17] 3GPP TS 24.237: "IP Multimedia Subsystem (IMS) Service Continuity; Stage 3".

[18] 3GPP TS 29.468: "Group Communication System Enablers for LTE (GCSE_LTE); MB2 Reference Point; Stage 3".

[19] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".

[20] 3GPP TS 24.008: "Mobile Radio Interface Layer 3 specification; Core Network Protocols; Stage 3".

[21] 3GPP TS 23.003: "Numbering, addressing and identification".

[22] ISO/IEC 9646-1: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[23] ISO/IEC 9646-7: "Information technology - Open systems interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

[24] 3GPP TS 23.303: "Proximity-based services (ProSe); Stage 2".

[25] IETF RFC 4566 (July 2006): "SDP: Session Description Protocol".

[26] 3GPP TS 26.171: "Speech codec speech processing functions; Adaptive Multi-Rate - Wideband (AMR-WB) speech codec; General description".

[27] 3GPP TS 36.523-3: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); User Equipment (UE) conformance specification; Part 3: Test suites".

[28] 3GPP TS 34.229-3: "Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 3: Abstract Test Suites (ATS)".

[29] ISO/IEC 9646-1: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[30] ISO/IEC 9646-7: "Information technology - Open systems interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

[31] ETSI ES 201 873-1: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 1: TTCN-3 Core Language".

[32] IETF RFC 3711: "The Secure Real-time Transport Protocol (SRTP)".

[33] 3GPP TS 27.007: "AT command set for User Equipment (UE)".

[34] IETF RFC 4661: "An Extensible Markup Language (XML)-Based Format for Event Notification Filtering".

[34] IETF RFC 4826: "Extensible Markup Language (XML) Formats for Representing Resource Lists".

[35] W3C: "XML Encryption Syntax and Processing Version 1.1", https://www.w3.org/TR/xmlenc-core1/.

[36] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[37] OMA - poc_listService-v1_0: "List Service".

[40] OMA - xdm_commonPolicy-V1_0: "XDM - Common Policy".

[39] OMA - xdm_extensions-v1_0: "XDM - XDM2 - Extensions".

[40] OMA - xdm_rsrclst_uriusage-v1_0: "Resource List - URI usage".

[41] W3C: "XML Encryption Syntax and Processing Version 1.1", https://www.w3.org/TR/xmlenc-core1/.

[42] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[43] 3GPP TS 33.180: "Security of the mission critical service".

[44] IETF RFC 6507: "Elliptic Curve-Based Certificateless Signatures for Identity-Based Encryption (ECCSI)".

[45] IETF RFC 6508: "Sakai-Kasahara Key Encryption (SAKKE)".

[46] IETF RFC 6509 (February 2012): "MIKEY-SAKKE: Sakai-Kasahara Key Encryption in Multimedia Internet KEYing (MIKEY)".

[47] IETF RFC 3394: "Advanced Encryption Standard (AES) Key Wrap Algorithm".

[48] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[49] IETF RFC 7515: "JSON Web Signature (JWS)".

[50] IETF RFC 5261: "An Extensible Markup Language (XML) Patch Operations Framework Utilizing XML Path Language (XPath) Selectors".

[51] IETF RFC 5874: "An Extensible Markup Language (XML) Document Format for Indicating a Change in XML Configuration Access Protocol (XCAP) Resources".

[52] IETF RFC 4354: "A Session Initiation Protocol (SIP) Event Package and Data Format for Various Settings in Support for the Push-to-Talk over Cellular (PoC) Service".

[53] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[54] 3GPP TS 23.280: "Common functional architecture to support mission critical services".

[55] 3GPP TS 24.281: "Mission Critical Video (MCVideo) signalling control; Protocol specification"

[56] 3GPP TS 24.581: "Mission Critical Video (MCVideo) media plane control; Protocol specification"

[57] 3GPP TS 24.282: "Mission Critical Data (MCData) signalling control; Protocol specification"

[58] 3GPP TS 24.582: "Mission Critical Data (MCData) media plane control; Protocol specification"

[59] 3GPP TS 37.579-6: "Mission Critical (MC) services; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification"

[60] 3GPP TS 37.579-7: "Mission Critical (MC) services; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification"

[61] 3GPP TS 36.508: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common Test Environments for User Equipment (UE) Conformance Testing".

[62] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace"

[63] IETF RFC 5245: "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols"

[64] IETF RFC 3830: "MIKEY: Multimedia Internet KEYing".

[65] IETF RFC 6509 (February 2012): "MIKEY-SAKKE: Sakai-Kasahara Key Encryption in Multimedia Internet KEYing (MIKEY)".

[66] IETF RFC 6043: "MIKEY-TICKET: Ticket-Based Modes of Key Distribution in Multimedia Internet KEYing (MIKEY)".

[67] IETF RFC 3550: "RTP: A Transport Protocol for Real-Time Applications".

[68] IETF RFC 4975: "The Message Session Relay Protocol (MSRP)".

[69] IETF RFC 2616: "Hypertext Transfer Protocol -- HTTP/1.1".

[70] ETSI ES 201 873-11: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 11: Using JSON with TTCN-3".

[71] OMA-TS-REST_NetAPI_NMS-V1_0-20190528-C: "RESTful Network API for Network Message Storage".

[72] 3GPP TS 38.523-3: "5GS; User Equipment (UE) conformance specification; Part 3: Protocol Test Suites".

[73] 3GPP TS 38.508-1: "5GS; User Equipment (UE) conformance specification; Part 1: Common test environment".

[74] 3GPP TS 38.508-2: "5GS; User Equipment (UE) conformance specification; Part 2: Common Implementation Conformance Statement (ICS) proforma".

[75] ETSI ES 201 873-9: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 9: Using XML schema with TTCN-3".

[76] OMA-TS-REST_NetAPI_NotificationChannel-V1_0-20200319-C: “RESTful Network API for Notification Channel”.

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

In addition for the purposes of the present document, the following terms, definitions, symbols and abbreviations apply:

- such given in ISO/IEC 9646-1 [22] and ISO/IEC 9646-7 [23]

NOTE: Some terms and abbreviations defined in [22] and [23] are explicitly included below with small modification to reflect the terminology used in 3GPP.

Implementation eXtra Information for Testing (IXIT): A statement made by a supplier or implementer of an UEUT which contains or references all of the information (in addition to that given in the ICS) related to the UEUT and its testing environment, which will enable the test laboratory to run an appropriate test suite against the UEUT.

IXIT proforma: A document, in the form of a questionnaire, which when completed for an UEUT becomes an IXIT.

Protocol Implementation Conformance Statement (PICS): An ICS for an implementation or system claimed to conform to a given protocol specification.

Protocol Implementation eXtra Information for Testing (PIXIT): An IXIT related to testing for conformance to a given protocol specification.

3.2 Symbols

No specific symbols have been identified so far.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

ASP	Abstract Service Primitive
ICS	Implementation Conformance Statement
IXIT	Implementation eXtra Information for Testing
MC	Mission Critical
MCData	Mission Critical Data
MCPTT	Mission Critical Push To Talk
MCS	Mission Critical Services
MCVideo	Mission Critical Video
PTC	Parallel Test Component
RTCP	RTP Control Protocol
RTP	Real-time Transport Protocol
SRTCP	Secure RTCP
SRTP	Secure RTP
SS	System Simulator
SSRC	Synchronization SouRCe
TC	Test Case
UE	User Equipment

4 Test system architecture

4.1 General system architecture

The architecture specified in TS 36.523-3 [27] clause 4.1.1 applies to the present document.

4.2 Component architecture

The architecture specified in TS 36.523-3 [27] clause 4.1.2 and TS 38.523-3 [72] clause 4.2 applies to the present document, with the clarification that only single RAT scenarios (E-UTRAN or NR/5GC) are considered by the test model of the present document.

5 Test models

5.1 MCX test model with TTCN based E-UTRA/EPC implementation (MCX EUTRA test model)

5.1.1 MCX Client on-network test model (EUTRA)

The MCX EUTRA on-network test model is depicted in figure 5.1.1-1. The test model consists of an IMS component and an HTTP component, on top of the multi-testers test model (E-UTRA) specified in TS 34.229-3 [28]. These parallel test components (PTCs) handle the IMS and HTTP signalling asynchronously.

The IMS PTC controls the IPCanEmu and the IP PTC. IPCanEmu is responsible for handling the E-UTRA cell(s) configuration in the SS as well as the E-UTRA/EPC level signalling and related procedures. The IPCanEmu and the NAS EMU are based on the TTCN implementation used for E-UTRA/EPC conformance testing according to TS 36.523-3 [27]. The IP PTC controls the IP related configurations as described in TS 36.523-3[27]. In addition, there is an SRTP port at the MCX IMS PTC mapped to the system interface to configure SRTP/SRTCP for media streams and media control messages (see clause 7.1.1.3). The media control messages are exchanged between TTCN and the system simulator via the IP PTC's IP_SOCK port (see clause 7.1.1.4 and clause 7.1.1.5) whereas the media stream messages are handled by the SS (see clause 7.1.1.5).

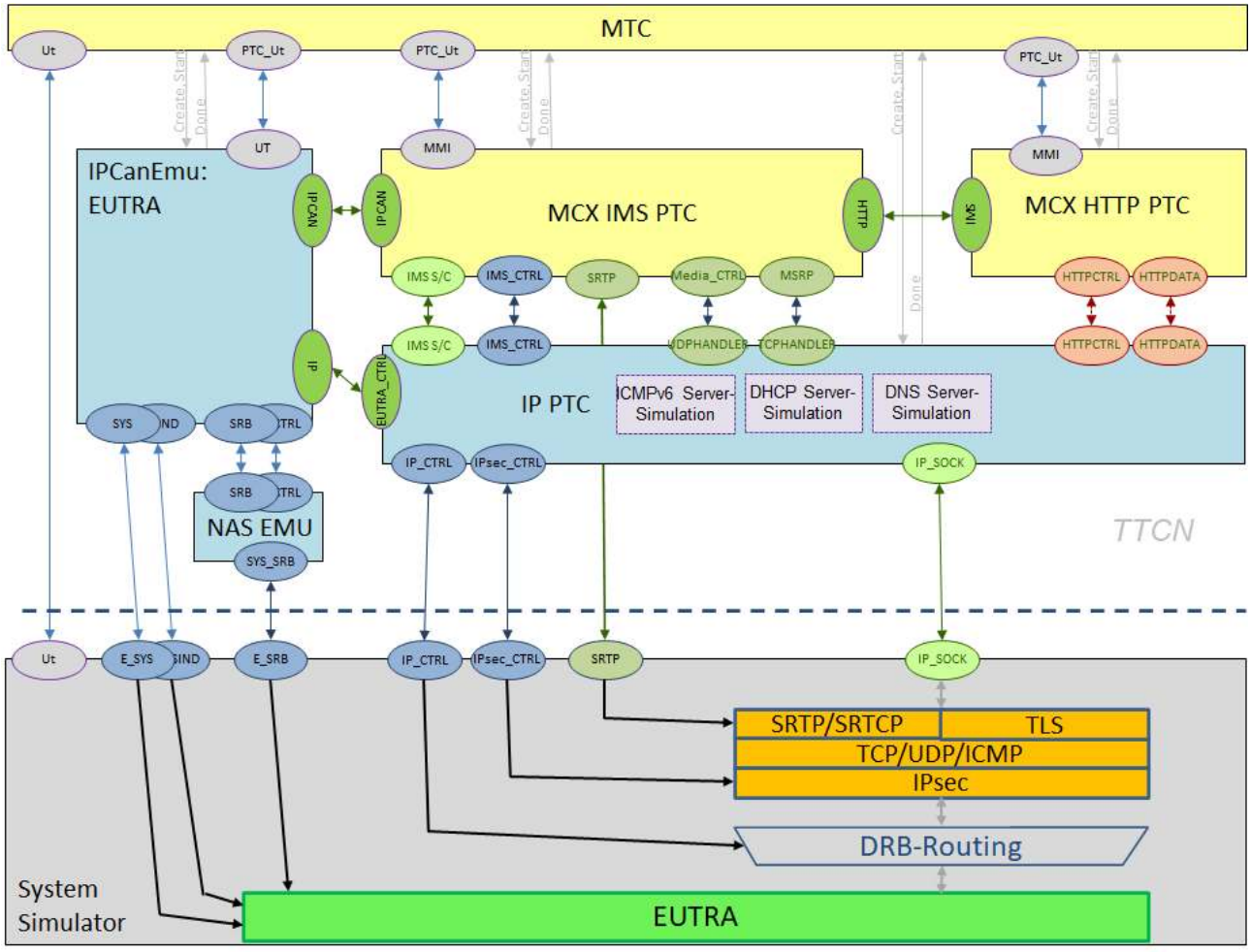


Figure 5.1.1-1: MCX Client on-network test model with TTCN based E-UTRA/EPC implementation

As described in TS 37.579-1 [2] clause 5.4.1A after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. Table 5.1.1-1 shows which IP addresses are assigned to a particular PDN.

Table 5.1.1-1: Mapping of IP addresses to PDNs

PDN	IP address	Value	Comment
MCX	NW IP address	px_IPv4_Address1_NW, px_IPv6_Address1_NW	PDN_1 in TTCN
	UE IP address	px_IPv4_Address1_UE, px_IPv6_Address1_UE	
IMS	NW IP address	px_IPv4_Address2_NW, px_IPv6_Address2_NW	PDN_2 in TTCN
	UE IP address	px_IPv4_Address2_UE, px_IPv6_Address2_UE	
Internet	NW IP address	px_IPv4_Address3_NW, px_IPv6_Address3_NW	PDN_3 in TTCN
	UE IP address	px_IPv4_Address3_UE, px_IPv6_Address3_UE	

5.1.2 MCX Client off-network test model (EUTRA)

An MCX Client off-network test model for EUTRA is not supported by the present version of the specification.

5.2 MCX test model with SS based E-UTRA/EPC or NR/5GC implementation (MCX IPCAN test model)

5.2.1 MCX Client on-network test model

In contrast to the MCX EUTRA test model (specified in clause 5.1.1) or NR5GC test model (specified in clause 5.3.1) the MCX IPCAN test model shown in figure 5.2.1-1 uses an external IPCAN (black-box) implementation which is integrated in the system simulator. In general, this IPCAN implementation shall be conformant in terms of TS 36.508 [61] and TS 38.508-1 [73] but this is out of the scope for the MCX IPCAN test model as long as the IPCAN implementation obeys the commands at the system interface as specified in annex E.

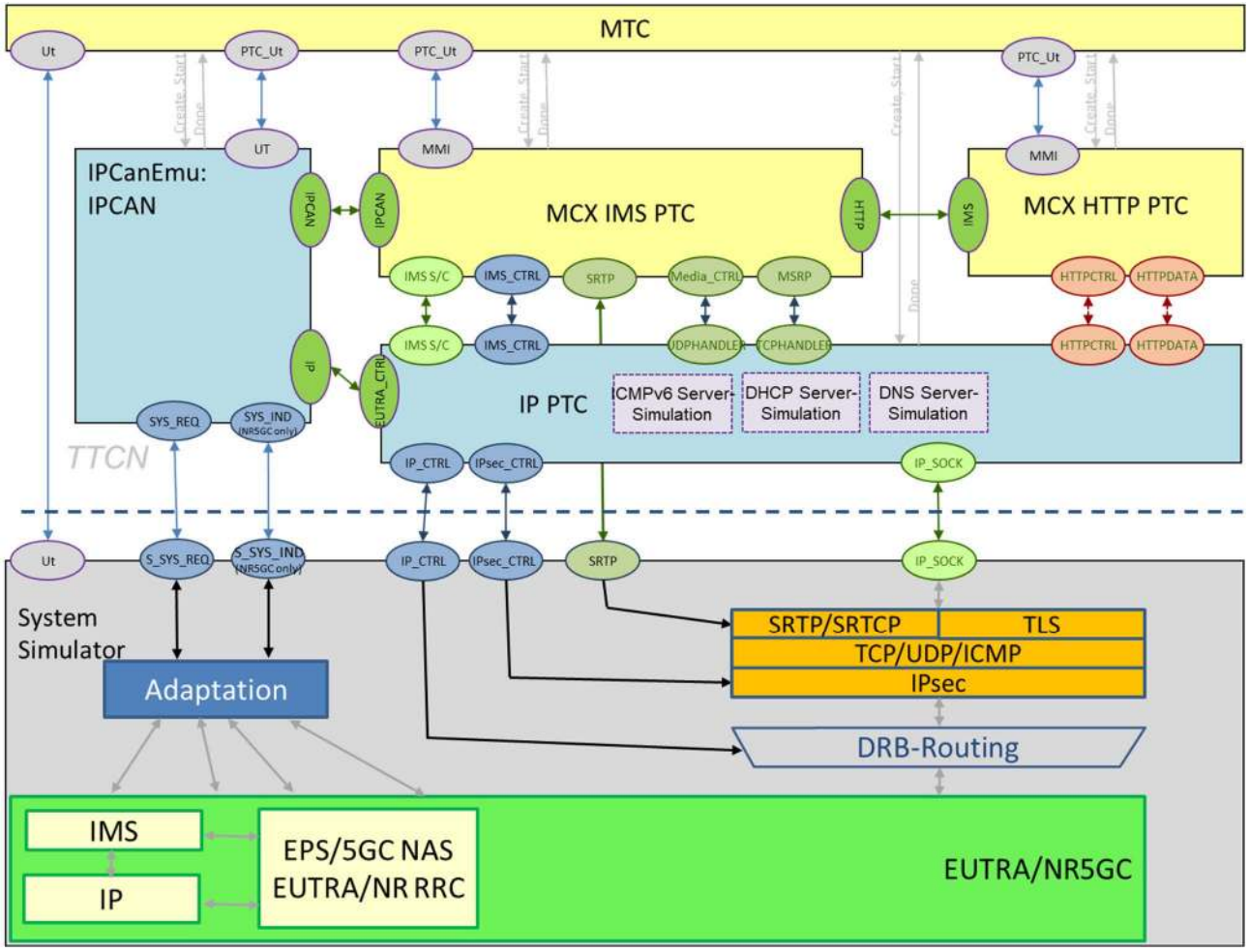


Figure 5.2.1-1: MCX Client on-network test model with SS based E-UTRA/EPC or NR/5GC implementation

NOTE 1: Whether or how the MCX IPCAN test model may use other IP-connectivity access networks is out of scope of this specification.

NOTE 2: The SYS_IND port shown in figure 5.2.1-1 is used only in case of NR5GC to indicate the PDU session id in a system indication and to assign the QFI in the corresponding acknowledgement.

NOTE 3: In the rest of this clause the term "PDN" is used for a PDN connectivity in EUTRA or a PDU session in NR5GC.

NOTE 4: The major procedures for configuration and control of the system simulator are illustrated in annex F.

As described in clauses 5.4.1A and 5.4.1B of TS 37.579-1 [2] after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. Nevertheless, in TTCN the MCX IPCAN test model supports signalling for the MCX PDN only: The IPCAN test model does not handle any signalling for the other PDNs and especially not the SIP signalling for an IMS registration to the IMS PDN.
⇒ It is up to SS implementation to handle the potential registrations to an internet PDN and/or an IMS PDN.

Tables 5.2.1-1 and 5.2.1-2 show the information provided to the SS for configuration of the PDNs for EUTRA and NR5GC.

Table 5.2.1-1: PDN info for EUTRA

PDN	Parameter	Value/Comment
MCX	EPS Bearer ID (NOTE 2)	px_MCX_IPCAN_EpsBearerId_MCX (NOTE 1)
	NW IP address	px_IPv4_Address1_NW, px_IPv6_Address1_NW (NOTE 3)
	UE IP address	px_IPv4_Address1_UE, px_IPv6_Address1_UE (NOTE 3)
IMS	EPS Bearer ID (NOTE 2)	px_MCX_IPCAN_EpsBearerId_IMS (NOTE 1)
	NW IP address	px_IPv4_Address2_NW, px_IPv6_Address2_NW
	UE IP address	px_IPv4_Address2_UE, px_IPv6_Address2_UE
Internet	EPS Bearer ID (NOTE 2)	px_MCX_IPCAN_EpsBearerId_INTERNET (NOTE 1)
	NW IP address	px_IPv4_Address3_NW, px_IPv6_Address3_NW
	UE IP address	px_IPv4_Address3_UE, px_IPv6_Address3_UE
NOTE 1: Values depending on SS implementation NOTE 2: The DRB associated with the EPS bearer shall have the DRB ID := EPS Bearer ID – 4. NOTE 3: Even though for MCX the user plane signalling is handled in TTCN still the SS needs the IP addresses to be provided to the UE in NAS signalling.		

Table 5.2.1-2: PDU session info for NR5GC

PDN	Parameter	Value/Comment
MCX	APN	pc_APN_ID_MCX (NOTE 2)
	NW IP address	px_IPv4_Address1_NW, px_IPv6_Address1_NW (NOTE 3)
	UE IP address	px_IPv4_Address1_UE, px_IPv6_Address1_UE (NOTE 3)
	QFI (NOTE 1)	11 (NOTE 4)
IMS	APN	pc_APN_ID_IMS (NOTE 2)
	NW IP address	px_IPv4_Address2_NW, px_IPv6_Address2_NW
	UE IP address	px_IPv4_Address2_UE, px_IPv6_Address2_UE
Internet	APN	pc_APN_ID_Internet (NOTE 2)
	NW IP address	px_IPv4_Address3_NW, px_IPv6_Address3_NW
	UE IP address	px_IPv4_Address3_UE, px_IPv6_Address3_UE
NOTE 1: Provided in acknowledgement of system indication indicating the PDU session id. NOTE 2: Defined in TS 38.508-2 [74] Table A.4.4-2A. NOTE 3: Even though for MCX the user plane signalling is handled in TTCN still the SS needs the IP addresses to be provided to the UE in NAS signalling. NOTE 4: According to TS 38.508-1 [73] Table 4.8.2.1-10 and Table 4.8.2.3-9.		

Dedicated EPS bearers and QoS flows use packet filters according to TS 36.508 [61] clause 6.6.2 and TS 38.508-1 [73] clause 4.8.2.2 as shown in table 5.2.1-3.

Table 5.2.1-3: Packet filters for dedicated EPS bearers and QoS flows

MC Service	Value/Comment		
MCPTT	Audio	Remote port range	tsc_MCX_MediaPort_Audio .. tsc_MCX_MediaPort_Audio + 1
		Protocol	UDP
	Media Control	Single remote port	tsc_MCX_MediaControlPortNW
		Protocol	UDP
MCVideo	Audio	Remote port range	tsc_MCX_MediaPort_Audio .. tsc_MCX_MediaPort_Audio + 1
		Protocol	UDP
	Video	Remote port range	tsc_MCX_MediaPort_Video .. tsc_MCX_MediaPort_Video + 1
		Protocol	UDP
	Media Control	Single remote port	tsc_MCX_MediaControlPortNW
		Protocol	UDP
MCData		Single remote port	tsc_MCData_MediaPort_MSRP
		Protocol	TCP

The respective packet filter is provided to the SS together with the EPS bearer id (EUTRA) or the QoS flow (NR5GC) at call establishment. EPS bearer ids and QoS flows are as shown in table 5.2.1-4.

Table 5.2.1-4: Dedicated EPS bearer ids and QoS flows

MC Service	EPS bearer id (EUTRA; NOTE 1)	QoS flow (NR5GC)	
		PDU session id	QFI
MCPTT	px_MCX_IPCAN_DedicatedEpsBearerId_MCPTT (NOTE 2)	(NOTE 3)	12 (NOTE 4)
MCVideo	px_MCX_IPCAN_DedicatedEpsBearerId_MCVideo (NOTE 2)	(NOTE 3)	13 (NOTE 4)
MCData	px_MCX_IPCAN_DedicatedEpsBearerId_MCData (NOTE 2)	(NOTE 3)	14 (NOTE 4)
NOTE 1: The DRB associated with the EPS bearer shall have the DRB ID := EPS Bearer ID – 4. NOTE 2: Values depending on SS implementation. NOTE 3: As chosen by the UE for the MCX PDU session at registration. NOTE 4: According to TS 38.508-1 [73] Table 4.8.2.1-11 and Table 4.8.2.3-10.			

5.3 MCX test model with TTCN based NR/5GC implementation (MCX NR5GC test model)

5.3.1 MCX Client on-network test model (NR5GC)

The MCX NR5GC on-network test model is depicted in figure 5.3.1-1.

The MCX NR5GC on-network test model is basically the same as described in clause 5.1.1 for E-UTRA with the following clarifications:

- In TTCN the NR5GC's IPCanEmu PTC and the NAS EMU NR PTCs replace the EUTRA's IPCanEmu and NAS EMU PTCs.
- The IPCanEmu and the NAS EMU NR are based on the TTCN implementation used for NR/5GC conformance testing according to TS 38.523-3 [72].
- At the system simulator an NR protocol stack is used instead of the E-UTRA protocol stack.

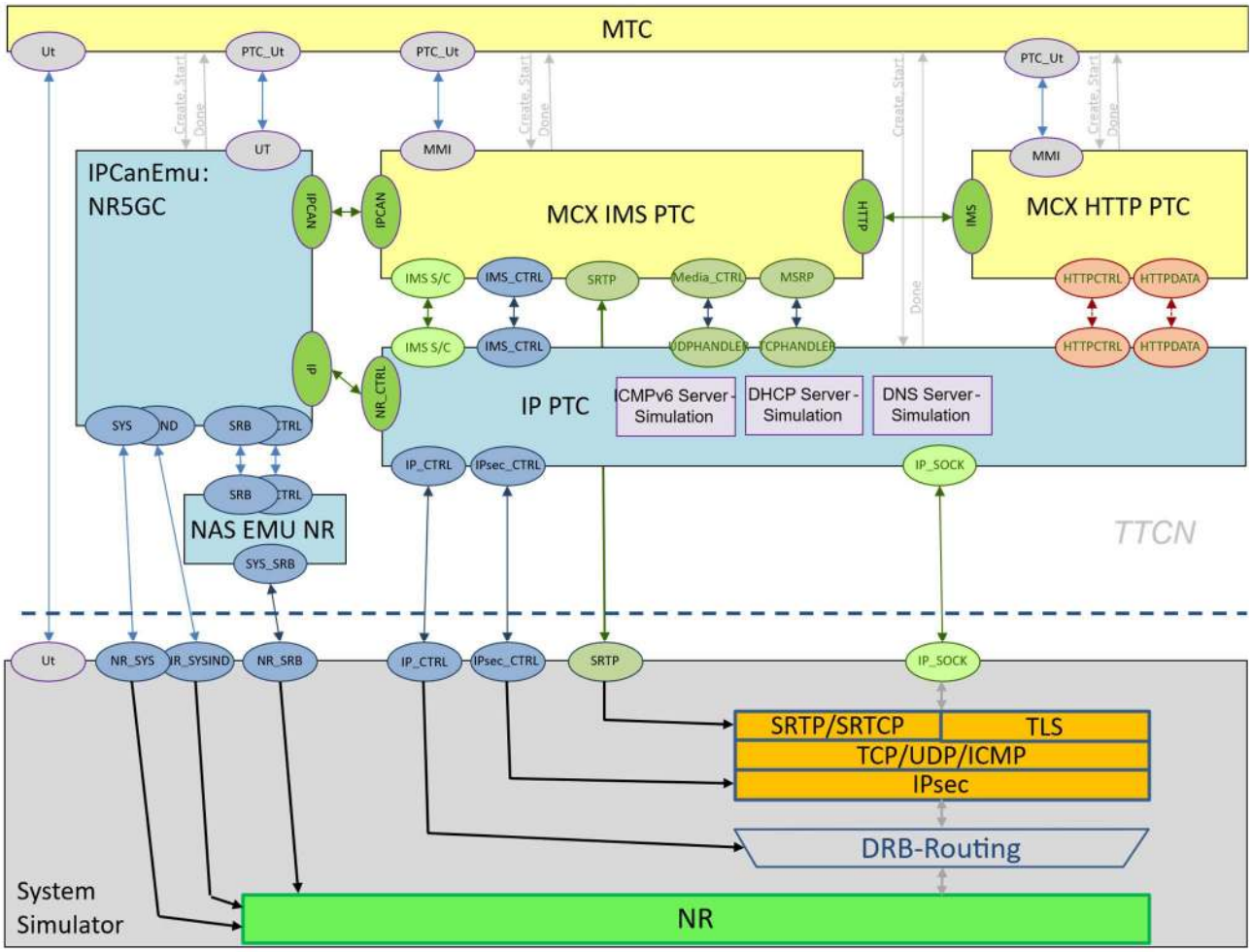


Figure 5.3.1-1: MCX Client on-network test model with TTCN based NR/5GC implementation

As described in TS 37.579-1 [2] clause 5.4.1B after switch on the UE may request an internet PDU session, an IMS PDU session and the MCX PDU session. Table 5.3.1-1 shows which IP addresses are assigned to a particular PDU session.

Table 5.3.1-1: Mapping of IP addresses to PDU sessions

PDU session	IP address	Value	Comment
MCX	NW IP address	px_IPv4_Address1_NW, px_IPv6_Address1_NW	PDN_1 in TTCN
	UE IP address	px_IPv4_Address1_UE, px_IPv6_Address1_UE	
IMS	NW IP address	px_IPv4_Address2_NW, px_IPv6_Address2_NW	PDN_2 in TTCN
	UE IP address	px_IPv4_Address2_UE, px_IPv6_Address2_UE	
Internet	NW IP address	px_IPv4_Address3_NW, px_IPv6_Address3_NW	PDN_3 in TTCN
	UE IP address	px_IPv4_Address3_UE, px_IPv6_Address3_UE	

5.3.2 MCX Client off-network test model (NR5GC)

An MCX Client off-network test model for NR/5GC is not supported by the present version of the specification.

6 System interface

6.1 Upper tester interface

The Upper Tester (UT) interface is the same as defined in TS 36.523-3 [27] clause 5. In addition there are the MCX specific MMI commands of table 6.1-1 and the MCX specific AT commands of table 6.1-2.

Table 6.1-1: MMI commands

Command	Parameters		ResultString
	Name	Value	
“MCX_USERACTION”	“UserAction”	<charstring>	(none)
“MCX_USERCHECK”	“UserCheck”	<charstring>	"YES" if the check is successful, "NO" otherwise

Table 6.1-2.: AT Commands

Command	Reference
AT+CAPTT	TS 27.007 [33]

6.2 Abstract system primitives

This clause specifies the abstract system primitives (ASPs) used on the system interface to configure and control the SS. The MCX test system interface re-uses the ASPs specified in TS 36.523-3 [27] (see clause 6 and Annex D), TS 38.523-3 [72] (see clause 6.2 and Annex D) and TS 34.229-3 [28] (see clause 6).

7 Test methods and design considerations

7.0 Introduction

The present document focusses on on-network client test cases for MCPTT (TS 37.579-2 [3]), MCVideo (TS 37.579-6 [59]) and MCData (TS 37.579-7 [60]). The major test behaviour and the verdict assignment of the test cases happens at the MCX IMS PTC and the MCX HTTP PTC as shown in the figures of clause 5.

7.1 Media plane signalling

7.1.0 Introduction

In addition to the HTTP and SIP signalling over control plane as used for the procedures in TS 24.379 [9], TS 24.281 [55] and TS 24.281 [57] there are MCX-specific procedures using the media plane as specified in TS 24.380 [10], TS 24.581 [56] and TS 24.581 [58].

7.1.1 MCPTT

7.1.1.1 Void

7.1.1.2 UDP/IP handling

The mechanisms specified in TS 36.523-3 [27] clause 4.2.4 are applied for UDP/IP configuration and Routing Table configuration.

7.1.1.3 RTP/RTCP handling

The RTP/RTCP loopback mechanism specified in TS 36.523-3 [27] clause 4.2.4.4 applies as baseline.

As MCPTT test cases require SRTP (secure RTP) for the media stream, the loopback mechanism needs to be enhanced: An RTP/RTCP packet received in uplink needs to be decrypted with the Rx or uplink key, encrypted with the Tx or downlink key and then sent back to the UE.

TTCN controls the usage of security for SRTP/SRTCP and provides the necessary security parameters to the SS via the SRTP port. Once security has been configured by TTCN, the SS shall handle media plane encryption and decryption.

The SRTP system interface is defined in Annex C.

7.1.1.4 Media Control handling

MCPTT requires that media control messages for floor control and pre-established session control according to TS 24.380 [10] are made available and handled in TTCN as structured messages.

TTCN uses the IP PTC to handle the UDP socket for media plane control messages: UDP packet information consisting of the encoded UDP payload and the local and remote socket information (IP address and port) is exchanged between the MCX IMS PTC and the IP PTC. The Client's port number is retrieved from the SDP signalling during connection establishment.

Media plane security is configured at the SRTP system port: The SS gets the information how to identify the UDP packets to be encrypted and decrypted together with the key material and the security policy parameters (see annex C).

7.1.1.5 SS pseudo-algorithm for RTP / RTCP / Media Plane Control handling

The following summarizes the requirements on the SS with regards to RTP / RTCP / Media Control handling within the SS:

- Uplink direction:
  - Upon reception of an SRTP/SRTCP packet, the SS shall extract the UDP payload and decrypt it using the RX crypto parameters of the respective connection.
  - In case of an RTP packet or an RTCP packet associated with the (RTP) media stream the SS shall apply the RTP/RTCP loopback as specified in clause 4.2.4.4 of TS 36.523-3 [27] with the following clarification:
    The SS shall replace the SSRC in the SRTP/SRTCP packet (octets 8 to 11 for SRTP and octets 4 to 7 for SRTCP, see IETF RFC 3711 [32] clauses 3.1 and 3.4) by the SSRC as configured at the SS for this connection.
    The SS shall encrypt the packet using the crypto parameters configured for the connection.
  - For RTCP APP packets the generic mechanisms of the IP test model according to clause 4.2 of TS 36.523-3 [27] apply: A UDP socket is configured by TTCN with corresponding entry in the common IP routing table so that the UDP packets are routed to the IP PTC which routes them to the MCX IMS PTC via the UDPHANDLER/Media_CTRL ports. Decoding of the media control messages is done in TTCN using TTCN-3's decvalue function.
- Downlink direction:
  - TTCN encodes media control messages using TTCN-3's encvalue function and sends the resulting datagram via the UDPHANDLER/Media_CTRL ports to the IP PTC and from there to the UDP socket configured at the SS. The SS shall encrypt the packet as according to the SRTP configuration for the connection and send it to the remote end.

7.1.2 MCVideo

The same principles apply as specified for MCPTT in clause 7.1.1 with the difference that the media plane control signalling is the transmission control protocol of TS 24.581 [56].

7.1.3 MCData

For MCData the Message Session Relay Protocol (MSRP) according to RFC 4975 [68] is used for test cases using the media plane. MSRP uses TCP as transport protocol.

The encoding, decoding and handling of the MSRP messages is done at the MCX IMS PTC which exchanges the encoded MSRP messages with the generic TCP handler of the IP PTC as shown the figures of clause 5.

The routing of the MSRP IP data is configured according to the IP Test Model of TS 36.523-3 [27] clause 4.2.4.

7.2 MCX functional architecture: HTTP, SIP

7.2.1 HTTP-Signalling

Figure 7.2.1-1 shows the functional architecture for HTTP-Signalling between the MCX-capable UE and different network entities as emulated by the test models.

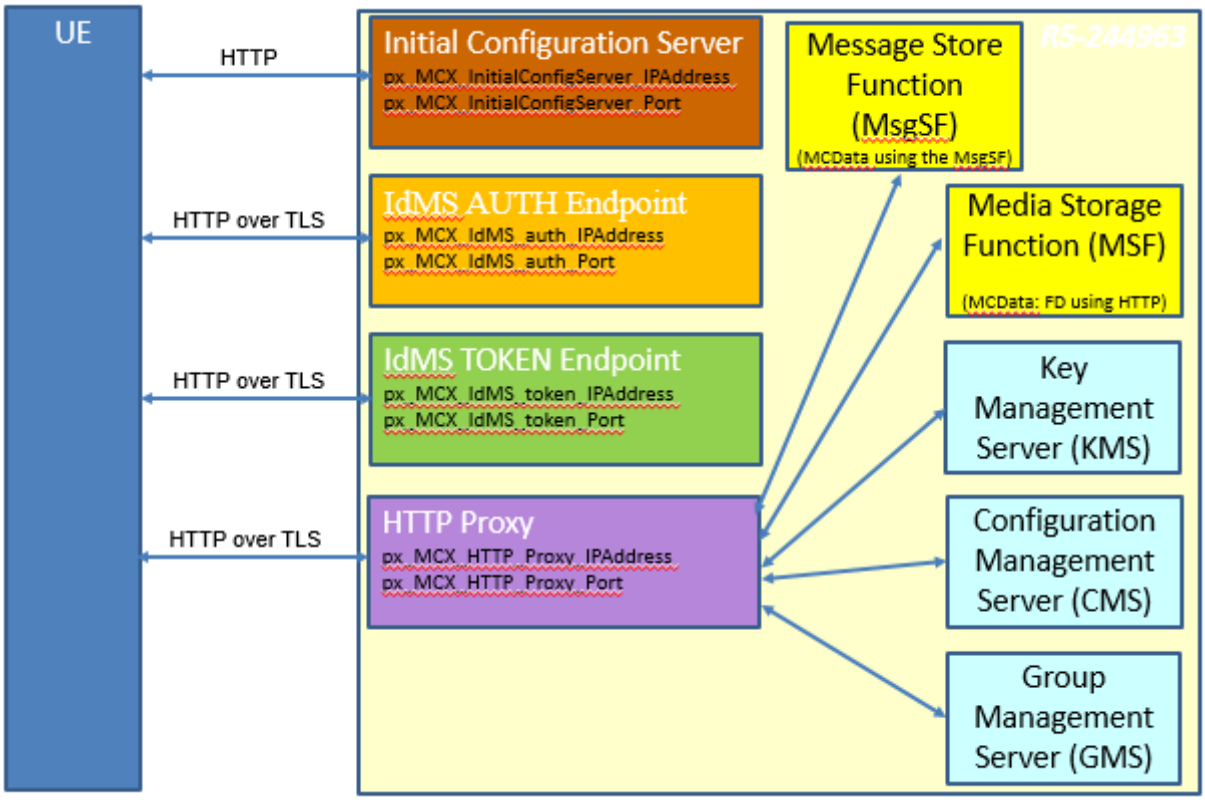


Figure 7.2.1-1: Functional architecture for HTTP-Signalling

The UE is expected to establish HTTP connections to the following entities:

- Initial Configuration Server:
  The UE gets configured with IP address (px_MCX_InitialConfigServer_IPAddress) and port number (px_MCX_InitialConfigServer_Port) of the server from which it retrieves the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  It depends on UE implementation how the IP address and port are configured at the UE but the UE is expected to support this to be configured.
  As there is no requirement by the core specs the HTTP connection is not security protected.

- IdMS AUTH Endpoint:
  The UE retrieves IP address (px_MCX_IdMS_auth_IPAddress) and port number (px_MCX_IdMS_auth_Port)

from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
The connection shall be TLS protected.

- IdMS TOKEN Endpoint:
  The UE retrieves IP address (px_MCX_IdMS_token_IPAddress) and port number (px_MCX_IdMS_token_Port) from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  The connection shall be TLS protected.

- HTTP Proxy:
  The UE retrieves IP address (px_MCX_HTTP_Proxy_IPAddress) and port number (px_MCX_HTTP_Proxy_Port) from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  The connection shall be TLS protected.

Design considerations and working assumptions:

- Domain Name Server
  DNS is not needed for address resolution as in the URIs of the network entities in the MCS UE initial configuration document the IP address of the respective entity can be used as host name.

- Initial UE configuration
  The UE is able to retrieve the MCS UE initial configuration document (TS 24.484 [14] clause 7.2) via an HTTP connection to an Initial Configuration Server. Other means of initial configuration are out of scope for conformance testing.

- HTTP signalling between UE and KMS
  The UE is expected to support HTTP signalling with the KMS via the HTTP proxy using a transport key (TrK) according to TS 33.180 [43] clause 4.3.3 and clause 5.3.2.

- Use of TLS
  All HTTP traffic between the UE and the HTTP proxy and between the UE and the IdMS is protected using HTTPS (TS 33.180 [43] clause 5.1.3.1, clause 6.1.1 and B.12).
  NOTE: Even though NOTE 2 of TS 24.482 [12] clause 6.2.1 might be interpreted as if the HTTP signalling to the authorization endpoint may not be TLS secured, 33.180 [43] mandates TLS to be used.

- IP Addresses and port numbers
  IP addresses and port numbers of the network entities as shown in figure 7.2.1-1 are given as PIXIT (see clause 9). As only restriction the tuple {IP address, port number} needs to be unique for each entity, but there is no restriction no to use one and the same IP address for all entities as long as they use different port numbers.

- According to TS 24.282 [57] clause 10.2.2.1 the HTTP signalling between the MCData client and the media storage function (MSF) happens via an HTTP proxy according to annex A of TS 24.482 [12]. It is assumed that this is the same proxy as used for signalling with the KMS, CMS and GMS.

- According to TS 24.282 [57] clause 21.1 the HTTP signalling between the MCData message store client and MCData message store function (MsgSF) happens via an HTTP proxy according to annex A of TS 24.482 [12]. It is assumed that this is the same proxy as used for signalling with the KMS, CMS and GMS.

- According to RFC 2616 [69] clause 5.1.2 HTTP requests via an HTTP proxy require the Request-URI to be an absolute URI ⇒ HTTP requests to KMS, CMS and GMS shall have an absolute URI as Request-URI.

7.2.2 SIP-Signalling

Figure 7.2.2-1 shows the functional architecture for SIP-Signalling between the MCX-capable UE and different network entities as emulated by the test model(s).

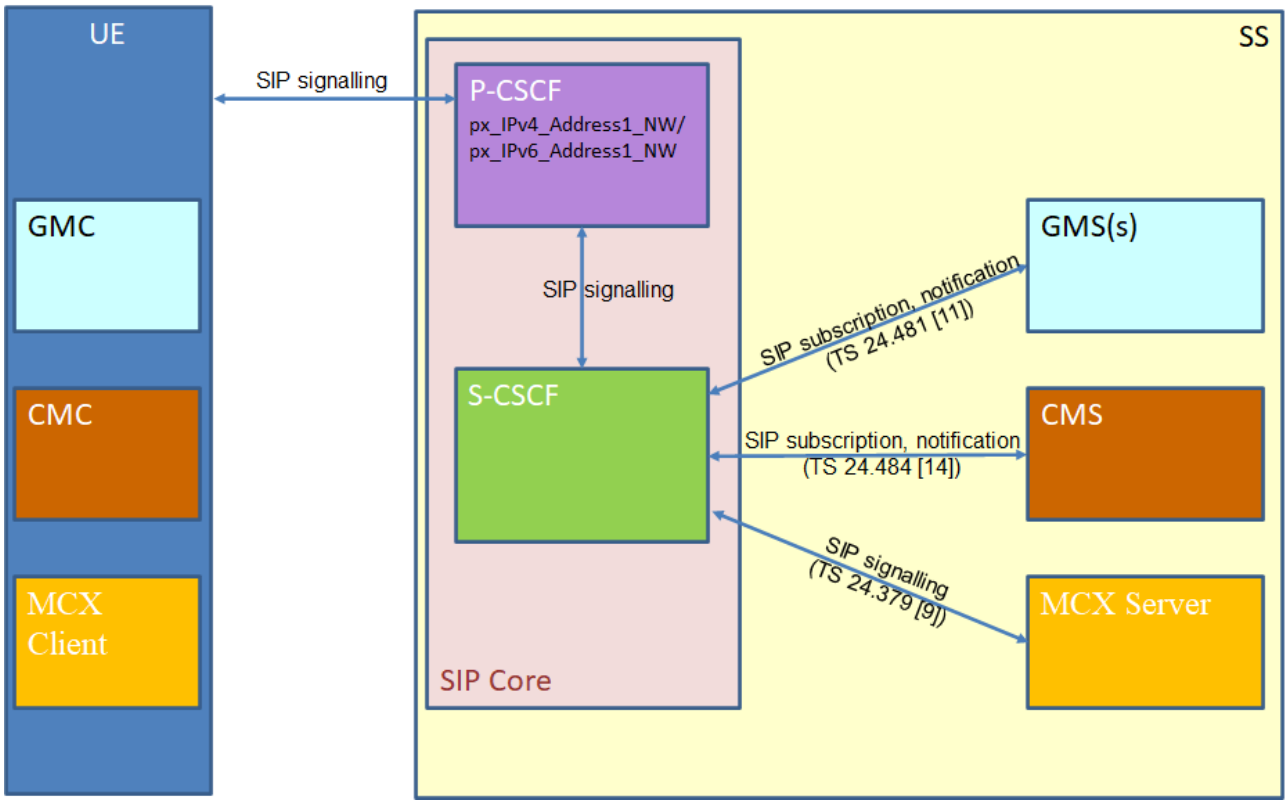


Figure 7.2.2-1: Functional architecture for SIP-Signalling

Regarding lower layer and IP signalling there are no differences compared to the IMS test model (TS 34.229-3 [28]).

The SIP signalling between the UE and the network entities is as specified in the test cases and according to common procedures and message content in TS 37.579-1 [2] with the following clarifications:

- Dialogs between the Group Management Client (GMC) and the Group Management Server (GMS)
  In general a GMC may subscribe to more than one GMS and TS 24.481 [11] shall be interpreted so that the GMC may even create more than one dialog with a single GMS.
  NOTE 1: In principle this applies for the Configuration Management Server too.
  NOTE 2: It is up to test case specification to deal with the possible scenarios and combinations of dialogs.

- Client Server Keys
  As not specified otherwise in the core specifications (TS 24.379 [9], TS 24.481 [11], TS 24.484 [14] and TS 33.180 [43]), there are no restriction for the UE regarding the CSK used in a dialog with the GMS(s), the CMS or the MCX server.
  ⇒ In general for CSK upload the UE may create and use different CSKs for the different servers and even for one and the same server the UE may use different CSKs for different dialogs. Furthermore a CSK download by the MCX server affects the communication between the MCX client and the MCX server only but not the CSKs used in any existing or future dialogs with a GMS or CMS; i.e. when the UE creates a dialog with a GMS or CMS after CSK download, it may use the downloaded CSK or any other CSK.

7.3 Interactive connectivity establishment (ICE) for pre-established session

According to TS 24.379 [9] for establishment of a pre-established session in general the MCPTT client shall gather ICE candidates according to RFC 5245 [63] and provide the candidates in the SDP offer of the initial SIP INVITE. Similar to the client the MCPTT server shall provide ICE candidates in its SDP answer and in general the client starts connectivity checks according to RFC 5245 [63].

Nevertheless, to reduce the test model requirements, the server side (at the system simulator) can be a lite implementation according to RFC 5245 [63] and as reflected in the default message contents for SDP Messages from the SS in TS 37.579-1 [2].

In addition - as working assumption - the UE shall be configured as lite implementation too.

⇒ When both ends are lite implementations none of them will start connectivity checks. Furthermore, there is no need for emulation for a STUN server for the client to gather ICE candidates as a lite implementation only includes host candidates for any media stream (RFC 5245 [63] clause 2.7).

Conclusion:
As long as there are no test requirements regarding ICE and as long as clients can be configured as lite implementations there are no ICE specific requirements for the test model.

8 Other SS requirements with TTCN-3 impact

8.1 Codec requirements

In addition to the requirements specified in TS 36.523-3 [27] clause 8.1, TS 38.523-3 [72] clause 8.1 and TS 34.229-3 [28] clause 7 the codec requirements of table 8.1-1 shall be applied.

Table 8.1-1: Codec requirements

Type definitions	Codec requirements	Encoding rule in TTCN-3
MIKEY_TypeDefs	RFC 3830 [64] clause 6 with extensions in RFC 6509 [65] and RFC 6043 [66]	"MIKEY-Codec"
MIKEY_MCX_Extension_TypeDefs	TS 33.180 [43] Annex E.6	"Key Parameter Payload Types"
MCDataMessage_TypeDefs	TS 24.282 [57] clause 15	"MCData Messages"
MCPTT_MediaControl_TypeDefs FloorControl_Group PreestablishedSessionControl_Group	 TS 24.380 [10] clause 8.2 TS 24.380 [10] clause 8.3	 "FloorCtrlCodec" "PreestablishedSessionCtrlCodec"
MCVideo_TransmissionControl_TypeDefs	TS 24.581 [56] clause 9	"TransmissionCtrlCodec"
JSON	ES 201 873-11 [70] annex B, B.2	"JSON"

XML schemas are used according to ETSI ES 201 873-9 [75] with clarification provided in annex D.1.

NOTE: For XML schemas, in general, type definitions of different name spaces are considered as not compatible, i.e. in TTCN there are no common templates for type definitions of different name spaces even when the converted TTCN types would be compatible according to clause 6.3 of ES 201 873-1 [31].

8.2 External function definitions

In addition to the external functions specified in TS 36.523-3 [27] clause 8.2 and TS 38.523-3 [72] clause 8.2 there are the following MCX specific external functions:

TTCN-3 External Function		
Name	**fx_SAKKE_GeneratePublicKey**	
Description	Generate KMS public key (Z_T) for SAKKE (RFC 6508 [45] clause 2.2): Z_T := [z_T]P	
Parameters	p_MasterSecret	master secret z_T (octetstring)
	p_ParameterSet	parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46]
Return Value	octetstring	

TTCN-3 External Function		
Name	**fx_SAKKE_GenerateRSK**	
Description	Generate receiver secret key (RSK) for SAKKE (RFC 6508 [45] clause 2.2): RSK := [(a + z_T)^-1]P with 'a' being the identifier (UID) corresponding to the receiver's URI	
Parameters	p_MasterSecret	master secret z_T (octetstring)
	p_Identifier	UID for a given URI (octetstring)
	p_ParameterSet	parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46]
Return Value	octetstring	

TTCN-3 External Function		
Name	**fx_SAKKE_EncapsulateKey**	
Description	Generate encapsulated data for SAKKE exchange according to RFC 6508 [45]	
Parameters	p_SSV	Shared secret value (octetstring): Key to be exchanged; according to 33.180 [43] E.1.1: The GMK, PCK, CSK and MuSiK shall be 16 octets in length
	p_SakkePublicKey	SAKKE public key (octetstring) generated with fx_SAKKE_GeneratePublicKey
	p_UID	UID (octetstring) generated for the receiving entity's URI (in general the same URI as in IDRr payload of the MIKEY message carrying the encapsulated data)
	p_ParameterSet	parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46]
Return Value	octetstring	

TTCN-3 External Function		
Name	**fx_SAKKE_ExtractKey**	
Description	Extract and validate 16 octet key from the encapsulated date for SAKKE exchange according to RFC 6508 [45] clause 6.2.2: if validation fails (TEST does not equal R_(b,S)) then omit shall be returned	
Parameters	p_EncapsulatedData	encapsulated data (octetstring) as received in the SAKKE payload of a MIKEY message
	p_SakkeRSK	receiver secret key (RSK) for SAKKE (octetstring)
	p_SakkePublicKey	SAKKE public key (octetstring) generated with fx_SAKKE_GeneratePublicKey
	p_UID	UID (octetstring) generated for the receiving entity's URI (should be the same URI as in IDRr payload of the MIKEY message carrying the encapsulated data
	p_ParameterSet	parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46]
Return Value	template (omit) octetstring (16 octets if the key valid, omit otherwise)	

TTCN-3 External Function		
Name	**fx_ECCSI_GenerateKPAK**	
Description	Generate KMS Public Authentication Key (KPAK) for ECCSI (RFC 6507 [44] clause 4.2): KPAK := [KSAK]G The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used.	
Parameters	p_KSAK	KMS Secret Authentication Key (KSAK): random secret non-zero integer modulo q (octetstring)
Return Value	octetstring	

TTCN-3 External Function		
Name	**fx_ECCSI_GenerateSskPvtPair**	
Description	Generate (SSK,PVT) Pair according to clause 5.1.1 of RFC 6507 [44]; if either the SSK or HS is zero modulo q (step 5 of the algorithm), the function shall return omit The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used.	
Parameters	p_UID	User ID (octetstring)
	p_KSAK	KMS Secret Authentication Key (KSAK) (octetstring)
	p_KPAK	KMS Public Authentication Key (KPAK) (octetstring)
	p_EphemeralValue	random (ephemeral) non-zero integer value v according to RFC 6507 [44] clause 5.1.1

Return Value	template (omit) type record ECCSI_SskPvtPair_Type {
	octetstring SSK,
	octetstring PVT
	}	

TTCN-3 External Function		
Name	**fx_ECCSI_SignMessage**	
Description	Sign a message according to RFC 6507 [44] clause 5.2.1: return signature of the message or omit if HE + r * SSK is non-zero modulo q (step 4 of the algorithm) The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used.	
Parameters	p_Message	Message to be signed (octetstring)
	p_KPAK	KMS Public Authentication Key (KPAK) (octetstring)
	p_UID	Signer's User ID (octetstring)
	p_SSK	Secret Signing Key (octetstring)
	p_PVT	Public Validation Token (octetstring)
	p_EphemeralValue	random (ephemeral) non-zero integer value j according to RFC 6507 [44] clause 5.2.1
Return Value	template (omit) octetstring	

TTCN-3 External Function		
Name	**fx_ECCSI_VerifySignature**	
Description	Verify a signature according to RFC 6507 [44] clause 5.2.2: return true, when the signature is valid, false otherwise The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used.	
Parameters	p_Message	Message (octetstring)
	p_Signature	Message's signature (octetstring)
	p_KPAK	KMS Public Authentication Key (KPAK) (octetstring)
	p_UID	Signer's User ID (octetstring)
Return Value	boolean	

TTCN-3 External Function		
Name	**fx_XML_Encrypt**	
Description	Encrypt data NOTE: the function is defined similar to openssl_encrypt and in principle it is independent from XML; nevertheless it is used for XML encrytion in context of MCX	
Parameters	p_Data	(octetstring)

	p_Method	type enumerated XML_EncryptionMethod_Type {
		AES_128_GCM, // AES-GCM according to clause 5.2.4 of
		// https://www.w3.org/TR/xmlenc-core1/ [41]
		// with 96 bit Initialization Vector and
		//128 bit Authentication Tag
		AES_256_KEY_WRAP // according to RFC 3394 [47]
		}

	p_Key	key for encryption (bitstring)
	p_IV	initial vector (octetstring)
Return Value	octetstring	In case of an authenticated encryption mechanism (e.g. AES_128_GCM) the Authentication Tag is added at the end of the encrypted data.

TTCN-3 External Function		
Name	**fx_XML_Decrypt**	
Description	Decrypt data	
Parameters	p_Data	In case of an authenticated encryption mechanism (e.g. AES_128_GCM) the input data is the concatination of the encrypted data and Authentication Tag (octetstring)

	p_Method	type enumerated XML_EncryptionMethod_Type {
		AES_128_GCM, // AES-GCM according to clause 5.2.4 of
		// https://www.w3.org/TR/xmlenc-core1/ [41]
		// with 96 bit Initialization Vector and
		//128 bit Authentication Tag
		AES_256_KEY_WRAP // according to RFC 3394 [47]
		}

	p_Key	key for encryption (bitstring)
	p_IV	initial vector (octetstring)
Return Value	octetstring	

TTCN-3 External Function		
Name	**fx_MCX_XML_AddEnvelopedSignature**	
Description	Add XML signature to the given XML document and return resulting XML document; according to W3C: "XML Signature Syntax and Processing (Second Edition)" [48]: 1. The given document has a Signature element with the name of the key to be used to sign the Signature's SignedInfo 2. Elements(s) to be signed: For elements which are addressed by a reference URI in the Signature's SignedInfo the DigestValue shall be generated and added to the corresponding Reference element of the Signature's SignedInfo 3. The SignedInfo shall be signed by generating the hash for the Signature's SignedInfo using the given key; this hash value shall be added to the Signature's SignatureValue.	
Parameters	p_XmlDocument	XML document (charstring) to be signed; the document contains all information to get signed with the given key: - id(s) for the object(s) to be signed (e.g. KMS response) - SignedInfo with reference(s) to objects to be signed within the document ("same-document" reference(s) according to RFC 3986 [53] clause 4.4), DigestAlgorithm, empty DigestValue
	p_Key	Key corresponding to the KeyName in the Signature's KeyInfo element (bitstring)
Return Value	universal charstring containing the document with all DigestValues and the signature of the SignedInfo	

TTCN-3 External Function		
Name	**fx_MCX_XML_CreateDetachedSignature**	
Description	Create XML signature for the given XML document and return resulting (encoded) XML signature (detached signature); according to TS 33.180 [43] clause 9.3.5, TS 24.379 [9] clause 6.6.3.1 and W3C: "XML Signature Syntax and Processing (Second Edition)" [48]: p_SignatureInfo contains the parameters needed to create the signature and to sign the document; the whole document gets signed irrespective of the Reference URI	
Parameters	p_XmlDocument	XML document to be signed
	p_XmldsigDocument	XML xmldsig document with signature element containing all information to sign the document (p_XmlDocument)
	p_Key	Key corresponding to the KeyName in the Signature's KeyInfo element
Return Value	universal charstring	XML xmldsig document with DigestValue and SignatureValue added to p_XmldsigDocument

TTCN-3 External Function		
Name	**fx_XML_GetElementContent**	
Description	Return content of the node in an XML tree or XML branch as pointed at by the xpath expression	
Parameters	p_XmlBranch	XML tree (including the root element) or branch ot an XML tree
	p_XpathExpression	XCAP expression according to https://www.w3.org/TR/1999/REC-xpath-19991116/
Return Value	universal charstring	string containing the content of the element or empty string if p_XpathExpression does not address an existing element of the given XML branch NOTE: if the element may be empty, it may be necessary to check existence of the element with fx_XML_GetElement first

TTCN-3 External Function		
Name	**fx_XML_SetElementContent**	
Description	Return XML tree or XML branch with replaced content of the node pointed at by the xpath expression	
Parameters	p_XmlBranch	XML tree (including the root element) or branch ot an XML tree
	p_XpathExpression	XCAP expression according to https://www.w3.org/TR/1999/REC-xpath-19991116/
	p_NewContent	new content for the given node
Return Value	universal charstring	modified XML tree or XML branch

TTCN-3 External Function		
Name	**fx_SHA_1**	
Description	Generic SHA-1 function	
Parameters	p_Data	data (octetstring)
Return Value	octetstring (hash value of 160 bits)	

TTCN-3 External Function		
Name	**fx_SHA_2**	
Description	Generic SHA-2 function	

Parameters	p_Function	type enumerated HASH_Function_Type {
		SHA_256
		// may be extended e.g SHA_224 etc.
		}

	p_Data	data (octetstring)
Return Value	octetstring (representing 256 bits for SHA-256, 224 bits for SHA-224, ...)	

TTCN-3 External Function		
Name	**fx_JWK_Signature**	
Description	Generate JWK signature according to RFC 7515 [49]	
Parameters	p_String	string for which the signature shall be generated (charstring)

	p_Algorithm	algorithm to generate the hash:
		
		type enumerated JWK_HashAlgorithm_Type { // RFC 7515 [49]
		HS256, // HMAC SHA-256
		RS256 // RSASSA-PKCS1-v1_5 SHA-256
		}

	p_Key	key (octetstring)
Return Value	charstring (base64url encoded signature according to RFC 7515 [49])	

TTCN-3 External Function		
Name	**fx_XML_Log**	
Description	Log function for (unencrypted) XML documents	
Parameters	p_FileName	TTCN module from where the function is called
	p_LineNo	Line number from where the function is called
	p_UplinkDownlink	XML document has been received (UPLINK) or will be sent (DOWNLINK)
	p_XmlDocument	XML document

TTCN-3 External Function		
Name	**fx_AEAD_Encrypt**	
Description	Encrypt data using AEAD algorithms according to RFC 5116	
Parameters	p_Data	(octetstring)
	p_AssociatedData	(octetstring)

	p_Method	type enumerated AEAD_EncryptionMethod_Type {
		AEAD_AES_128_GCM, // RFC 5116 clause 5.1
		AEAD_AES_256_GCM // RFC 5116 clause 5.2
		}

	p_Key	key for encryption (bitstring)
	p_IV	initial vector (octetstring)
Return Value	octetstring	Encrypted data appended by authentication tag

TTCN-3 External Function		
Name	**fx_AEAD_Decrypt**	
Description	Decrypt data using AEAD algorithms according to RFC 5116	
Parameters	p_Data	(octetstring)
	p_AssociatedData	(octetstring)

	p_Method	type enumerated AEAD_EncryptionMethod_Type {
		AEAD_AES_128_GCM, // RFC 5116 clause 5.1
		AEAD_AES_256_GCM // RFC 5116 clause 5.2
		}

	p_Key	key for decryption (bitstring)
	p_IV	initial vector (octetstring)
Return Value	octetstring	Decrypted data

9 IXIT Proforma

9.1 General

This partial IXIT proforma contained in the present document is provided for completion, when the related Abstract Test Suite is to be used against the Implementation Under Test (IUT).

Text in italics is a comment for guidance for the production of an IXIT, and is not to be included in the actual IXIT.

The completed partial IXIT will normally be used in conjunction with the completed ICS, as it adds precision to the information provided by the ICS.

9.2 MCX

9.2.1 MCX Client PIXIT

Table 9.2.1-1: MCX Client Common PIXIT

Parameter Name	Parameter Type	Default Value	Supported Values	Description
Client relevant IXIT				
px_MCX_Client_B_ID	charstring	"urn:uuid:cdcd34fb-c2f1-4488-a03a-6b5a5d360c47"		MCX client ID for a remote user: As defined in TS 24.379 [9] clause 4.10 it is a UUID URN according to RFC 4122 [62], i.e. has the format "urn:uuid:XXXXXXXX-YYYY-ZZZZ-yyyy-zzzzzzzzzzzz"
Users relevant IXIT				
px_MCPTT_ID_User_A	charstring	"mcptt-user-A-id@mcptt-op.gov"	NOTE 1	MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13].
px_MCVideo_ID_User_A	charstring		NOTE 1	MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13].
px_MCData_ID_User_A	charstring		NOTE 1	MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13].
px_MCPTT_User_A_Alias	charstring	"mcptt-user-A-alias"		Alphanumeric alias of MCPTT user. Ref. TS 24.483 [13].
px_MCVideo_User_A_Alias	charstring	"mcvideo-user-A-alias"		Alphanumeric alias of MCVideo user. Ref. TS 24.483 [13].
px_MCData_User_A_Alias	charstring	"mcdata-user-A-alias"		Alphanumeric alias of MCData user. Ref. TS 24.483 [13].
px_MCPTT_ID_FA_A	charstring	"mcptt-FA-A-id@mcptt-op.gov"		The value is a "uri" attribute. TS 24.483 [13] clause 5.2.48W6
px_MCData_ID_FA_A	charstring	"mcdata-FA-A-id@mcptt-op.gov"		The value is a "uri" attribute. TS 24.483 [13] clause 10.2.97B3
px_MCVideo_ID_FA_A	charstring	"mcvideo-FA-A-id@ mcptt-op.gov"		The value is a "uri" attribute. TS 24.483 [13] clause 13.2.87A6
px_MCX_User_A_ParticipantType	charstring	"first responder"		Participant type of the MCPTT user. Ref. TS 24.483 [13].
px_MCX_DomainName_Organization_A	charstring	"mcptt-op.gov"		Domain name of the organization the user belongs to.
px_MCX_User_A_username	charstring	"MCPTT#U01"		UE's User username used for user authentication
px_MCX_User_A_password	charstring	"psw@MCPTT&7"		UE's User password used for user authentication
px_MCX_TokenRsaPrivateKey	octetstring			Private key being used to create signature for ID and Access Token
px_MCX_TrK	bitstring			Transport key for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3.
px_MCX_TrK_ID	B32_Type			Transport key ID for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3.
px_MCX_InK	bitstring			Integrity protection key for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3.

Parameter Name	Parameter Type	Default Value	Supported Values	Description
px_MCX_InK_ID	B32_Type			Integrity protection key ID for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3.
px_MCPTT_ID_User_B	charstring	"mcptt-user-B-id@ mcptt-op.gov"		MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13].
px_MCVideo_ID_User_B	charstring	"mcvideo-user-B-id@ mcptt-op.gov"		MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13].
px_MCData_ID_User_B	charstring	"mcdata-user-B-id@ mcptt-op.gov"		MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13].
px_MCX_User_B_ParticipantType	charstring	"first responder"		Participant type of the MCX user. Ref. TS 24.483 [13].
px_MCPTT_ID_User_C	charstring	"mcptt-user-C-id@ mcptt-op.gov"		MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13].
px_MCVideo_ID_User_C	charstring	"mcvideo-user-C-id@ mcptt-op.gov"		MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13].
px_MCData_ID_User_C	charstring	"mcdata-user-C-id@ mcptt-op.gov"		MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13].
px_MCX_User_C_ParticipantType	charstring	"first responder"		Participant type of the MCX user. Ref. TS 24.483 [13].
px_MCPTT_ID_User_D	charstring	"mcptt-user-D-id@ mcptt-op.gov"		MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13].
px_MCVideo_ID_User_D	charstring	"mcvideo-user-D-id@ mcptt-op.gov"		MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13].
px_MCData_ID_User_D	charstring	"mcdata-user-D-id@ mcptt-op.gov"		MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13].
px_MCX_SIP_PublicUserId_A_1	charstring			(First) public user id of the SIP subscriber (user A) for MCX; the parameter shall be a SIP URI

Parameter Name	Parameter Type	Default Value	Supported Values	Description
px_MCX_SIP_PrivateUserId_A	charstring			Private user id of the SIP subscriber (user A) for MCX
px_MCX_SIP_PrivateUserId_B	charstring			Private user id of the SIP subscriber (user B) for MCX
px_MCX_SIP_HomeDomain_A	charstring			Home domain name of the SIP subscriber (user A) for MCX
px_MCX_SIP_PublicUserId_B	charstring			Public user id of user B
px_MCX_SIP_PublicUserId_C	charstring			Public user id of user C
px_MCX_SIP_RegistrationWithTemporaryIdentities	boolean			If true the UE shall derive the public user id, private user id and home domain name from the IMSI for SIP registration
px_MCData_IPConnectivityEndpointAddress_B	charstring			IP address of the IP connectivity endpoint at the emulated client B: either an IPv4 or an IPv6 address
Groups relevant IXIT				
px_MCPTT_Group_A_ID	charstring	"mcptt-group-A@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCVideo_Group_A_ID	charstring	"mcvideo-group-A@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCData_Group_A_ID	charstring	"mcdata-group-A@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCPTT_Group_A_Name	charstring	"mcptt-group-A-name"		A human readable Group name for the group
px_MCVideo_Group_A_Name	charstring	"mcvideo-group-A-name"		A human readable Group name for the group
px_MCData_Group_A_Name	charstring	"mcdata-group-A-name"		A human readable Group name for the group
px_MCX_Group_A_Owner_Organization	charstring	"mcx-op.gov"		Indicates the group's owner organization the group belongs to. Ref. TS 24.483 [13].
px_MCPTT_Group_A_preferred_VCodec	charstring	"AMR-WB"		Preferred voice codec for the group (a RTP payload). MCPTT clients shall support the AMR-WB codec. RFC 4566 [25] TS 26.171 [26]
px_MCPTT_Group_B_ID	charstring	"mcptt-group-B@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCVideo_Group_B_ID	charstring	"mcvideo-group-B@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCData_Group_B_ID	charstring	"mcdata-group-B@mcptt-op.gov"		Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].

Parameter Name	Parameter Type	Default Value	Supported Values	Description
px_MCPTT_Group_T_ID	charstring	"mcptt-group-T@mcptt-op.gov"		Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCVideo_Group_T_ID	charstring	"mcvideo-group-T@mcptt-op.gov"		Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
px_MCData_Group_T_ID	charstring	"mcdata-group-T@mcptt-op.gov"		Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13].
Miscellaneous IXIT				
px_MCX_APN	charstring	"mcx-apn"		A single APN which the UE shall use to access each and all MCX relevant services. The APN is provided in the initial UE configuration as specified in TS 37.579-1 [2] Table 5.5.8.1-1.
px_MCX_InitialRegistration_TypeOf1stPDN	MCX_Registration_PDN_Type	mcx	ims, internet, mcx	First PDN registered during initial registration (either 'ims' or 'internet' or 'mcx'; 'none' is not applicable as first PDN). (NOTE 2)
px_MCX_InitialRegistration_TypeOf2ndPDN	MCX_Registration_PDN_Type	none	ims, internet, mcx, none	Second PDN registered during initial registration; in addition to 'ims' or 'internet' or 'mcx' it may be 'none' to indicate that there is no second PDN connectivity requested by the UE during initial registration. (NOTE 2)
px_MCX_InitialRegistration_TypeOf3rdPDN	MCX_Registration_PDN_Type	none	ims, internet, mcx, none	Third PDN registered during initial registration; in addition to 'ims' or 'internet' or 'mcx' it may be 'none' to indicate that there is no third PDN connectivity requested by the UE during initial registration. (NOTE 2)
px_MCX_CoordinateLatitude_Client_B	float	none		Simulated latitude value sent to the UE to inform about Client_B location
px_MCX_CoordinateLongitude_Client_B	float	none		Simulated longitude value sent to the UE to inform about Client_B location
px_MCX_CiphAlgo_Def	CiphAlgo	des_ede3_cbc	des_ede3_cbc, aes_cbc or nociph	Ciphering Algorithm for IPsec. NOTE: Unless specified otherwise in the test prose „nociph" shall not be used for verification.
px_MCX_IPSecAlgorithm	IntAlgo	hmac_sha_1_96	hmac_md5_96, hmac_sha_1_96	Integrity Algorithm for IPsec

NOTE 1: According to TS 23.280 [54] clause 8.1.2 a MC service ID shall be a URI; nevertheless in context of this specification only URIs consisting of userinfo, host and path elements (see RFC 3986 [53]) are supported.
Editor's note: A similar note may need to be added to other PIXITs too.

NOTE 2: Basically, there is no correlation to the TTCN PDN_Index_Type specifying PDN_1, PDN_2, etc. The mapping depends on the TTCN implementation for the respective IPCAN.

9.2.2 MCX Server PIXIT

Table 9.2.2-1: MCX Server Common PIXIT

Parameter Name	Parameter Type	Default Value	Supported Values	Description
px_MCX_GroupCreationXUI	charstring	"mcx-gms@mcptt-op.gov"		Indicates the group creation XUI information for creation of groups. Ref. TS 23.003 [21].
px_MCX_IdMS_auth_IPAddress	charstring			IPv4/IPv6 address of the IdMS authorization endpoint
px_MCX_IdMS_auth_Port	integer			Port number of the IdMS authorization endpoint
px_MCX_IdMS_auth_Certificate	charstring			Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the IdMS authorization endpoint
px_MCX_IdMS_token_IPAddress	charstring			IPv4/IPv6 address of the IdMS token endpoint
px_MCX_IdMS_token_Port	integer			Port number of the IdMS token endpoint
px_MCX_IdMS_token_Certificate	charstring			Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the IdMS token endpoint
px_MCX_HTTP_Proxy_IPAddress	charstring			IPv4/IPv6 address of the HTTP proxy
px_MCX_HTTP_Proxy_Port	integer			Port number of the HTTP proxy
px_MCX_HTTP_Proxy_Certificate	charstring			Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the HTTP proxy
px_MCX_InitialConfigServer_IPAddress	charstring			IPv4/IPv6 address of the server optionally being used to download the Initial UE Configuration document
px_MCX_InitialConfigServer_Port	integer			Port number of the server optionally being used to download the Initial UE Configuration document
px_MCX_InitialConfigServer_UriPath	charstring	"/cms/initial-ue-config"		URI Path component: Absolute path used for HTTP requests addressing a server to download the UE initial configuration document
px_MCX_TLS_CipherSuite	TLS_CIPHER_Type	TLS_RSA_WITH_NULL_SHA		Cipher suite to be used for TLS connections
px_MCX_OAuth_ClientId_A	charstring			Client ID of the UE's MCX application as used in OAuth signalling with the IdMS
px_MCX_OAuth_RedirectURI_A	charstring			Redirect URI used by the UE implementation (user agent, MCX client and OS) to redirect the OAuth authentication response to the MCX client application

9.2.3 MCX Constant PIXIT Definitions

Several parameters for MCX conformance testing can be defined as constants as they are neither preconfigured at the UE nor at the SS. Table 9.2.3-1 lists these constants.

Table 9.2.3-1: MCX Constant PIXIT Definitions

Constant Name	Constant Type	Value	Description
tsc_MCX_KMS_Hostname	charstring	"kms." & px_MCX_DomainName_Organization_A	FQDN of the KMS; used in initial UE configuration as domain name for the 'kms' URI in the App-Server-Info.
tsc_MCX_CMS_Hostname	charstring	"cms." & px_MCX_DomainName_Organization_A	FQDN of the CMS; used in initial UE configuration as domain name for the 'cms' URI in the App-Server-Info.
tsc_MCX_GMS_Hostname	charstring	"gms." & px_MCX_DomainName_Organization_A	FQDN of the GMS; used in initial UE configuration as domain name for the 'gms' URI in the App-Server-Info.
tsc_MCX_KMS_CertUri	charstring	"certificate1." & tsc_MCX_KMS_Hostname	Name of the KMS certificate sent to the UE during MCX user authentication
tsc_MCX_IdMS_auth_UriPath	charstring	"/idms/auth"	URI path to address the IdMS authorization endpoint
tsc_MCX_IdMS_token_UriPath	charstring	"/idms/token"	URI path to address the IdMS token endpoint
tsc_MCX_IdMS_userauth_UriPath	charstring	"/idms/userauth"	relative URI at IdMS used for user authentication
tsc_MCX_KMS_init_UriPath	charstring	"/keymanagement/identity/v1/init"	Path of the Request-URI for "KMS Initialize" request according to TS 33.180 [43] D.2.3
tsc_MCX_KMS_keyprov_UriPath	charstring	"/keymanagement/identity/v1/keyprov"	Path of the Request-URI for "KMS KeyProvision" request according to TS 33.180 [43] D.2.4
tsc_MCX_GMSURI	charstring	"subscription-proxy." & tsc_MCX_GMS_Hostname	used for <GMS-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9; sub-domain of the GMS's hostname
tsc_MCX_CMSXCAPRootURI	charstring	"http://xcap." & tsc_MCX_CMS_Hostname	used for <CMS-XCAP-root-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9C
tsc_MCX_GMSXCAPRootURI	charstring	"http://xcap." & tsc_MCX_GMS_Hostname	used for <GMS-XCAP-root-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9B
tsc_MCPTT_PublicServiceId_A	charstring	"sip:" & "mcptt-server-A@" & px_MCX_DomainName_Organization_A	The URI of the MCPTT Server which is simulated by the SS
tsc_MCVideo_PublicServiceId_A	charstring	"sip:" & "mcvideo-server-A@" & px_MCX_DomainName_Organization_A	The URI of the MCVideo Server which is simulated by the SS
tsc_MCData_PublicServiceId_A	charstring	"sip:" & "mcdata-server-A@" & px_MCX_DomainName_Organization_A	The URI of the MCData Server which is simulated by the SS

Constant Name	Constant Type	Value	Description
tsc_MCPTT_PublicServiceId_B	charstring	"sip:" & "mcptt-server-B@" & px_MCX_DomainName_Organization_A	The URI of a second MCPTT Server which is implemented in the DUT used in MCPTT Server testing.
tsc_MCPTT_PublicServiceId_PF_A	charstring	"sip:" & "participating-mcptt-function-A@" & px_MCX_DomainName_Organization_A	The URI of the participating MCPTT function which configures the location reporting at the UE according to TS 24.379 [9] clause 13.2.2.
tsc_MCVideo_PublicServiceId_PF_A	charstring	"sip:" & "participating-mcvideo-function-A@" & px_MCX_DomainName_Organization_A	The URI of the participating MCVideo function which configures the location reporting at the UE according to TS 24.281 [55] clause 18.2.2.
tsc_MCData_PublicServiceId_PF_A	charstring	"sip:" & "participating-mcdata-function-A@" & px_MCX_DomainName_Organization_A	The URI of the participating MCData function which configures the location reporting at the UE according to TS 24.282 [57] clause 17.2.2.
tsc_MCX_SessionID_B	charstring	"sip:" & "sessionB@cf." & px_MCX_DomainName_Organization_A	The URI of the MCPTT session B identity. Ref. TS 24.483 [13]. SIP URI according to TS 24.379 [9] clause 4.5.
tsc_MCPTT_SessionId	charstring	"sip:" & "mcptt-session-A@cf." & px_MCX_DomainName_Organization_A	Service Id for MCPTT calls as used as Contact URI of the MCPTT server
tsc_MCVideo_SessionId	charstring	"sip:" & "mcvideo-session-A@cf." & px_MCX_DomainName_Organization_A	Service Id for MCVideo calls as used as Contact URI of the MCVideo server
tsc_MCData_SessionId	charstring	"sip:" & "mcdata-session-A@cf." & px_MCX_DomainName_Organization_A	Service Id for MCData calls as used as Contact URI of the MCData server
tsc_MCData_MSF_Hostname	charstring	"msf." & px_MCX_DomainName_Organization_A	Hostname of the media storage function (MSF)
tsc_MCData_MSF_URI	charstring	"http://" & tsc_MCData_MSF_Hostname & "/userA/files"	Absolute URI of the media storage function (MSF)
tsc_MCData_MsgSF_Hostname	charstring	"msgsf." & px_MCX_DomainName_Organization_A	Hostname of the message store function (MsgSF)
tsc_MCData_MsgSF_BaseURL	charstring	"/MsgStoreRoot/nms/v1/mcc160/UserA"	Base URL of the message store function (MsgSF) according to clause 5.2 of RESTful Network API for NMS [71]
tsc_MCData_MsgSF_AbsoluteBaseURL	charstring	"http://" & tsc_MCData_MsgSF_Hostname & tsc_MCData_MsgSF_BaseURL	Absolute Base URL of the message store function (MsgSF)
tsc_MCPTT_Group_A_ProSeLayer2GroupID	octetstring	'00000A'O	Indicates the Prose layer-2 group ID for the group. Ref. TS 23.303 [24].
tsc_MCPTT_Group_D_ProSeLayer2GroupID	octetstring	'00000D'O	Indicates the Prose layer-2 group ID for the group. Ref. TS 23.303 [24].
tsc_MCVideo_VSF_Hostname	charstring	"vsf." & px_MCX_DomainName_Organization_A	Hostname of the video storage function (VSF) where video files are stored

Constant Name	Constant Type	Value	Description
tsc_MCVideo_VSF_URI	charstring	"http://" & tsc_MCVideo_VSF_Hostname & "/userA/files"	Absolute URI of the message storage function (MSF) where video files are stored
tsc_MCData_NS_Hostname	charstring	"ns." & px_MCX_DomainName_Organization_A	Hostname of the notification server (NS)
tsc_MCData_NS_BaseURL	charstring	"/NSserverRoot/notificationchannel/v1"	Base URL of the notification server (NS) according to clause 5.2 of RESTful Network API for NC [76]

9.3 MCX IPCAN test model

Table 9.3-1: MCX IPCAN test model specific PIXIT

Parameter Name	Parameter Type	Default Value	Supported Values	Description
px_MCX_IPCAN_EpsBearerId_IMS	MCX_IPCAN_EpsBearerId_Type	5	5..13	Default EPS bearer for IMS PDN (if any)
px_MCX_IPCAN_EpsBearerId_INTERNET	MCX_IPCAN_EpsBearerId_Type	6	5..13	Default EPS bearer for Internet PDN (if any)
px_MCX_IPCAN_EpsBearerId_MCX	MCX_IPCAN_EpsBearerId_Type	7	5..13	Default EPS bearer for MCX PDN
px_MCX_IPCAN_DedicatedEpsBearerId_MCPTT	MCX_IPCAN_EpsBearerId_Type	8	5..13	Dedicated EPS bearer for MCPTT
px_MCX_IPCAN_DedicatedEpsBearerId_MCVideo	MCX_IPCAN_EpsBearerId_Type	9	5..13	Dedicated EPS bearer for MCVideo
px_MCX_IPCAN_DedicatedEpsBearerId_MCData	MCX_IPCAN_EpsBearerId_Type	10	5..13	Dedicated EPS bearer for MCData

10 Postambles

10.1 Introduction

The purpose of the present clause 10 is to specify the postambles used to bring the UE to a well-defined state regardless of the UE state at the termination of main test body or of the SS conditions and values of the system information inherited from the test.

10.2 MCX

The postambles specified in TS 34.229-3 [28] are also applicable to MCX test cases.

Annex A (normative): Test Suites

A.1 Introduction

This annex references the approved Test Suites, which accompany the present document. The Test Suites have been produced using the Testing and Test Control Notation version 3 (TTCN-3) according to ES 201 873-1 [31].

A.2 Baseline of specifications

Table A.2-1 lists the core specifications and test specifications, which the delivered Test Suites are based upon.

Table A.2-1: References of the test and Core specifications

Type	Specification	Release	Version
Core specifications	TS 24.379 [9]	Note 1	Note 2
	TS 24.380 [10]	Note 1	Note 2
	TS 24.481 [11]	Note 1	Note 2
	TS 24.482 [12]	Note 1	Note 2
	TS 24.483 [13]	Note 1	Note 2
	TS 24.484 [14]	Note 1	Note 2
	TS 33.180 [43]	Note 1	Note 2
	TS 24.229 [16]	Note 1	Note 2
	TS 24.281 [55]	Note 1	Note 2
	TS 24.581 [56]	Note 1	Note 2
	TS 24.282 [57]	Note 1	Note 2
	TS 24.582 [58]	Note 1	Note 2
Test specifications	TS 37.579-1 [2]	Note 1	Note 2
	TS 37.579-2 [3]	Note 1	Note 2
	TS 37.579-4 [5]	Note 1	Note 2
	TS 37.579-6 [59]	Note 1	Note 2
	TS 37.579-7 [60]	Note 1	Note 2
NOTE 1: Latest release available, up to the release number of the present document. NOTE 2: Latest available			

A.3 MCX Client Test Suites

A.3.1 MCPTT Client Test Suites

Table A.3.1-1 lists all approved test cases from TS 37.579-2 [3].

For a given test case, the following variants are distinguished (if applicable):

- IPCAN 4G: MCX IPCAN 4G test model.
- EUTRA: MCX EUTRA test model.
- IPCAN 5G: MCX IPCAN 5G test model.
- NR5GC: MCX NR5GC test model.

An "X" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is approved for the respective variant.

An "-" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is not applicable to the respective variant.

Table A.3.1-1: MCPTT Client TTCN test cases from TS 37.579-2 [3]

Test case	Description	IPCAN 4G	EUTRA	IPCAN 5G	NR5G
1.MCPTT	MCPTT / Configuration / Authentication / User Authorisation / UE Configuration / User Profile	X	X		X
3.MCPTT	MCPTT / Configuration / Group Affiliation / Remote change / De-affiliation / Home MCPTT system	X			
4.MCPTT	MCPTT / Configuration / Pre-established Session Establishment / Pre-established Session Modification / Pre-established Session Release	X			
6.MCPTT	Configuration / Download CSK	X			
8.MCPTT	Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias	X			
1.1.1.MCPTT	MCPTT / On-network / On-demand Pre-arranged Group Call / Automatic Commencement Mode / Floor Control / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group Call / Cancel Imminent Peril State / Client Originated (CO)	X			
1.1.2.MCPTT	MCPTT / On-network / On-demand Pre-arranged Group Call / Automatic Commencement Mode / Floor Control / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group Call / Cancel Imminent Peril State / Client Terminated (CT)	X			
1.1.3.MCPTT	MCPTT / On-network / On-demand Pre-arranged Group Call / Manual Commencement Mode / Client Originated (CO)	X			
1.1.4.MCPTT	MCPTT / On-network / On-demand Pre-arranged Group Call / Manual Commencement Mode / Client Terminated (CT)	X			
1.1.5.MCPTT	MCPTT / On-network / Pre-arranged Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Originated (CO)	X			
1.1.6.MCPTT	MCPTT / On-network / Pre-arranged Group Call using pre-established session / Automatic Commencement Mode / Server originated Pre-established Session Release with associated MCPTT session / Client Terminated (CT)	X			
1.1.8.MCPTT	MCPTT / On-network / Pre-arranged Broadcast Group Call / Client Originated (CO)	X			
1.1.9.MCPTT	MCPTT / On-network / Pre-arranged Broadcast Group Call / Client Terminated (CT)	X			
1.1.10.MCPTT	MCPTT / On-network / Broadcast Group Call with Temporary Group / Client Originated (CO)	X			
1.1.11.MCPTT	MCPTT / On-network / Pre-arranged Emergency Group Call / Client Originated (CO)	X			
1.1.12.MCPTT	MCPTT / On-network / Pre-arranged Emergency Group Call / Client Terminated (CT)	X			
1.1.13.MCPTT	MCPTT / On-network / Pre-Arranged Imminent Peril Group Call / Client Originated (CO)	X			
1.1.14.MCPTT	MCPTT / On-network / Pre-Arranged Imminent Peril Group Call / Client Terminated (CT)	X			
1.1.16.MCPTT	MCPTT / On-network / Emergency Alert / Client Terminated (CT)	X	X		X
1.1.17.MCPTT	On-network / Broadcast Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Originated (CO)	X			
1.1.18.MCPTT	MCPTT / On-network / Broadcast Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Terminated (CT)	X			
1.1.19.MCPTT	On-network / On-demand Pre-arranged Group Call / Active functional alias / Client Originated (CO)	X			
1.1.20.MCPTT	On-network / On-demand Pre-arranged Group Call / Multi Talker	X			
1.1.21.MCPTT	On-network / On-demand Pre-arranged Group Call / No Implicit Floor Control / Client Originated (CO)	X			
1.2.7.MCPTT	MCPTT / On-network / Chat Group Call / Emergency Group Call / Client Originated (CO)	X			
1.2.8.MCPTT	MCPTT / On-network / Chat Group Call / Emergency Group Call / Client Terminated (CT)	X			
1.2.9.MCPTT	MCPTT / On-network / Chat Group Call / Imminent Peril Group Call / Client Originated (CO)	X			
1.2.10.MCPTT	MCPTT / On-network / Chat Group Call / Imminent Peril Group Call / Client Terminated (CT)	X			
1.2.11.MCPTT	MCPTT / On-network / Chat Group Call / Join Chat Group Session / Upgrade to Emergency / Cancel Emergency / Upgrade to Imminent Peril / Cancel Imminent Peril / Client Originated (CO)	X			
1.2.12.MCPTT	MCPTT / On-network / Chat Group Call / Upgrade to Emergency / Cancel Emergency / Upgrade to Imminent Peril / Cancel Imminent Peril / Client Originated (CT)	X			
1.2.13.MCPTT	On-network / Chat Group Call / Join Chat Group Session / Active functional alias / Client Originated (CO)	X			

2.1.MCPTT	MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / With Floor Control confidentiality and integrity protection/ Upgrade to Emergency Call / Cancellation of Emergency on User request / Client Originated (CO)	X			
2.2.MCPTT	MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / With Floor Control confidentiality and integrity protection/ Upgrade to Emergency Call / Cancellation of Emergency on User request / Client Terminated (CT)	X			
2.3.MCPTT	MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / Without Floor Control / Client Originated (CO)	X			
2.4.MCPTT	MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / Without Floor Control / Client Terminated (CT)	X			
2.5.MCPTT	MCPTT / On-network / Private Call / Emergency Private Call / On-demand / Automatic Commencement Mode / Force of automatic commencement mode / Without Floor Control / Client Originated (CO)	X			
2.6.MCPTT	MCPTT / On-network / Private Call / Emergency Private Call / On-demand / Manual Commencement Mode / Force of automatic commencement mode / Without Floor Control / Client Terminated (CT)	X			
2.7.MCPTT	MCPTT / On-network / Private Call / On-demand / Manual Commencement Mode / Without Floor Control / Client Originated (CO)	X			
2.8.MCPTT	MCPTT / On-network / Private Call / On-demand / Manual Commencement Mode / Without Floor Control / Client Terminated (CT)	X			
2.14.MCPTT	On-network / Private Call / Ambient listening call / Remotely initiated Ambient listening call / Remotely initiated ambient listening call release / Success / Client Originated (CO) / Server initiated ambient call release	X			
2.15.MCPTT	MCPTT / On-network / Private Call / Ambient listening call / Remotely initiated Ambient listening call / Remotely initiated ambient listening call release / Success / Client Terminated (CT)	X			
2.16.MCPTT	MCPTT / On-network / Private Call / Ambient listening call / Locally initiated Ambient listening call / Locally initiated ambient listening call release / Success / Client Originated (CO) / Server initiated ambient call release	X			
2.17.MCPTT	MCPTT / On-network / Private Call / Ambient listening call / Locally initiated Ambient listening call / Locally initiated ambient listening call release / Success / Client Terminated (CT)	X			

A.3.2 MCVideo Client Test Suites

None.

A.3.3 MCData Client Test Suites

Table A.3.3-1 lists all approved test cases from TS 37.579-7 [60].

For a given test case, the following variants are distinguished (if applicable):

- IPCAN 4G: MCX IPCAN 4G test model.
- EUTRA: MCX EUTRA test model.
- IPCAN 5G: MCX IPCAN 5G test model.
- NR5GC: MCX NR5GC test model.

An "X" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is approved for the respective variant.

An "-" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is not applicable to the respective variant.

Table A.3.3-1: MCData Client TTCN test cases from TS 37.579-7 [60]

Test case	Description	IPCAN 4G	EUTRA	IPCAN 5G	NR5GC
5.1.MCData	Configuration / Authentication / User Authorization / UE Configuration / User Profile / Key Generation	X			
6.1.1.MCData	On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / One-to-one Standalone SDS / Client Originated (CO)	X			
6.1.3.MCData	On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / Group Standalone SDS / Client Originated (CO)	X			
6.1.4.MCData	On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / Group Standalone SDS / Client Terminated (CT)	X			

Annex B (informative): Style Guide

B.1 Introduction

The style guide specified in TS 36.523-3 [27] Annex B applies to the present document.

Annex C (normative): Common TTCN-3 Definitions

C.0 Introduction

The present Annex C specifies the TTCN-3 type definitions used at the system interface to configure and control the SS. The type definitions are common, independent from test model.

In case of discrepancy between the content of the present Annex C and the equivalent TTCN-3 definitions / semantic requirements found in the TTCN modules provided as attachments to the present specification, the latter shall take precedence.

NOTE: This annex is automatically generated from the TTCN-3 modules provided as attachment to the present specification and containing the listed TTCN-3 type definitions.

C.1 SRTP_ASP_TypeDefs

C.1.1 Type_Definitions

SRTP_EncryptionAlgorithm_Type

TTCN-3 Enumerated Type	
Name	**SRTP_EncryptionAlgorithm_Type**
Comment	
Null	RFC 3830 clause 6.10.1
AES_CM	RFC 3830 clause 6.10.1
AES_F8	RFC 3830 clause 6.10.1
AES_GCM	RFC 7714 clause 14.3

SRTP_AuthenticationAlgorithm_Type

TTCN-3 Enumerated Type	
Name	**SRTP_AuthenticationAlgorithm_Type**
Comment	
Null	RFC 3830 clause 6.10.1
HMAC_SHA_1	RFC 3830 clause 6.10.1
RCCm1	RFC 4771 clause 4
RCCm2	RFC 4771 clause 4
RCCm3	RFC 4771 clause 4

SRTP_PRF_Type

TTCN-3 Enumerated Type	
Name	**SRTP_PRF_Type**
Comment	
AES_CM	RFC 3830 clause 6.10.1

SRTP_SecurityProfile_Type

TTCN-3 Record Type			
Name	**SRTP_SecurityProfile_Type**		
Comment	parameters for SRTP security profiles according to 33.180 annex E: - Table E.2.2-1 (RTP group call: GMK) - Table E.3.2-1 (RTP private call: PCK) - Table E.4.2-1 (RTCP: CSK and MuSiK)): The parameters may be contained/negotiated in MIKEY Security Policy Payload (RFC 3830)		
EncryptionAlgorithm	SRTP_EncryptionAlgorithm_Type		SRTP type 0; RFC 3830 clause 6.10.1
SessionEncryptionKeyLength	integer		SRTP type 1; RFC 3830 clause 6.10.1
AuthenticationAlgorithm	SRTP_AuthenticationAlgorithm_Type	opt	SRTP type 2; RFC 3830 clause 6.10.1
SessionSaltKeyLength	integer		SRTP type 4; RFC 3830 clause 6.10.1
PRF	SRTP_PRF_Type		SRTP type 5; RFC 3830 clause 6.10.1
KeyDerivationRate	integer		SRTP type 6; RFC 3830 clause 6.10.1
ROC_TransmissionRate	integer	opt	SRTP type 13; RFC 4771 clause 4
SRTP_AuthenticationTagLength	integer	opt	SRTP type 18; RFC 4771 clause 4
SRTCP_AuthenticationTagLength	integer	opt	SRTP type 19; RFC 4771 clause 4
AEAD_AuthenticationTagLength	integer		SRTP type 20; RFC 7714 clause 14.3

SRTP_KeyInfo_Type

TTCN-3 Record Type			
Name	**SRTP_KeyInfo_Type**		
Comment	SRTP Master Key, SRTP Master Salt and MKI according to 33.180 clauses 7.4.1, 7.4.2 and 9.4.6, RFC 3830 and RFC 3711		
MasterKey	octetstring		SRTP master key according to RFC 3711 (128 bits)
MasterSalt	octetstring		SRTP master salt according to RFC 3711 and RFC 7714 (96 bits): 33.180 clauses 7.5 and 9.4.8 specify the AEAD_AES_128_GCM algorithm as defined in RFC 7714 to be used for confidentiality and data authentication of the media stream and the media control data. RFC 7714 specifies in clause 12 a master salt length of 96 and in clause 11 that the AES_CM PRF KDF according to RFC 3711 shall be used. Nevertheless RFC 3711 clause 8.2 requires a master salt length of 112. This contradiction is resolved by the RFC Errata for RFC 7714: The master salt shall be 96 bit in the first place (as according to RFC 7714) but then multiplied by 2^16 to fit to the requirements of RFC 3711 (112 bit). => MasterSalt provides a 96-bit value which needs to be padded by the system adaptor to get the 112-bit master salt as needed according to RFC 3711
MKI	octetstring	opt	SRTP Master Key Identifier: if not present => no MKI to be added to the SRTP/SRTCP packet

SRTP_Config_Type

TTCN-3 Record Type			
Name	**SRTP_Config_Type**		
Comment	SRTP configuration parameters		
KeyInfo	SRTP_KeyInfo_Type		
SecurityProfile	SRTP_SecurityProfile_Type		

SRTP_Connection_Type

TTCN-3 Record Type			
Name	**SRTP_Connection_Type**		
Comment	configuration parameters for a single (UDP) connection		
Connection	IP_Connection_Type		RTP/RTCP configuration shall be applied for packets matching the given Connection: in general a connection is given by Protocol := udp Local := { IP address of the SS, port number at the SS } Remote := omit (but other matching criteria may be used if needed) NOTE: TTCN shall ensure that different connection can be identified without ambiguity (in general even the local port number should be sufficient)
LocalSSRC	O4_Type		SSRC of the SS
SRTP_Config	SRTP_Config_Type	opt	configuration of SRTP/SRTCP for the given connection; if not present => no SRTP/SRTCP

SRTP_ConnectionTable_Type

TTCN-3 Record of Type	
Name	**SRTP_ConnectionTable_Type**
Comment	
record of SRTP_Connection_Type	

SRTP_CTRL_REQ

TTCN-3 Union Type		
Name	**SRTP_CTRL_REQ**	
Comment	configuration ASP at the system interface	
Config	SRTP_ConnectionTable_Type	table of all connection to be configured for SRTP/SRTCP (or RTP/RTCP); the configuration overrides any previous configuration; it is not foreseen to add, modify or remove single connections

SRTP_CTRL_IND

TTCN-3 Union Type		
Name	**SRTP_CTRL_IND**	
Comment	ASP at the system interface for system confirmations and indications	
Confirm	Null_Type	confirmation for configuration
Error	Null_Type	indication of errors

C.1.2 System_Interface

SRTP_PORT

TTCN-3 Port Type		
Name	**SRTP_PORT**	
Comment		
out	SRTP_CTRL_REQ	
in	SRTP_CTRL_IND	

C.2 IP_ASP_TypeDefs

General Notes:
NOTE 1:
In general the handling of IP data shall be independent from the RAT being used on lower layers.
NOTE 2:
It shall be possible for SS implementation to reuse existing IP stack implementations in the system adaptor;
therefore the well-known concept of socket programming shall be supported
(regardless of whether those are used in the system adaptor implementation or not)
NOTE 3:
Since in general at the network side there are several different IP addresses the SS needs to simulate more than one IP address;
that can be based on a concept of multiple virtual network adaptors
NOTE 4:
There is no easy way to control the routing of IP data for an IP connection from above the IP stack
i.e. there are no parameters at the socket interface to determine e.g. cell id and DRB id
=> another independent logical entity (DRB-MUX) is needed below the IP stack which is responsible to control the routing of IP packets from/to DRBs in different cells of different RATs

Reference:
An introduction to socket programming can be found in
UNIX Network Programming Volume 1, Third Edition: The Sockets Networking API
by W. Richard Stevens, Bill Fenner, Andrew M. Rudoff

C.2.1 IP_Common

IP_Common: Basic Type Definitions

TTCN-3 Basic Types		
PortNumber_Type	UInt16_Type	

IPv4_AddrInfo_Type

TTCN-3 Record Type			
Name	**IPv4_AddrInfo_Type**		
Comment	IPv4 specific info of the socket addr (AF_INET)		
Addr	charstring		IP Address as string (IP v4 dot notation) to be converted to 32-bit unsigned integer

IPv6_AddrInfo_Type

TTCN-3 Record Type			
Name	**IPv6_AddrInfo_Type**		
Comment	IPv6 specific info of the socket addr (AF_INET6); NOTE: sin6_flowinfo can be ignored and set to 0		
Addr	charstring		to be converted to sin6_addr
ScopeId	UInt32_Type	opt	sin6_scope_id in general an IPv6 address is like "fe80::1%eth0" with eth0 being the network adaptor mapped to a scope id (Unix) assumption: for UE conformance testing it is not necessary to distinguish different scopes and the scope id in general can be determined by the system adaptor => omit

IP_AddrInfo_Type

TTCN-3 Union Type		
Name	**IP_AddrInfo_Type**	
Comment		
V4	IPv4_AddrInfo_Type	
V6	IPv6_AddrInfo_Type	

IP_Socket_Type

TTCN-3 Record Type			
Name	**IP_Socket_Type**		
Comment	Socket		
IpAddr	IP_AddrInfo_Type	opt	IP address
Port	PortNumber_Type	opt	port number

InternetProtocol_Type

TTCN-3 Enumerated Type	
Name	**InternetProtocol_Type**
Comment	
udp	
tcp	
icmp	
icmpv6	

IP_Connection_Type

TTCN-3 Record Type			
Name	**IP_Connection_Type**		
Comment	A connection between peer-to-peer entities is unambiguously defined by the protocol (udp/tcp/icmp/icmpv4), the local socket and the remote socket		
Protocol	InternetProtocol_Type		
Local	IP_Socket_Type	opt	
Remote	IP_Socket_Type	opt	

C.3 CommonDefs

CommonDefs: Constant Definitions

TTCN-3 Basic Types			
tsc_UInt16Max	integer	65535	
tsc_UInt32Max	integer	4294967295	

CommonDefs: Basic Type Definitions

TTCN-3 Basic Types		
O4_Type	octetstring length(4)	
Null_Type	boolean (true)	dummy type for 'typeless' fields in unions
UInt16_Type	integer (0 .. tsc_UInt16Max)	
UInt32_Type	integer (0 .. tsc_UInt32Max)	

C.4 References to TTCN-3

References to TTCN-3		
SRTP_ASP_TypeDefs	Common/SRTP_ASP_TypeDefs.ttcn	Rev 29933
IP_ASP_TypeDefs	IP_PTC/IP_ASP_TypeDefs.ttcn	Rev 27511
CommonDefs	Common/CommonDefs.ttcn	Rev 29871

Annex D (normative): SIP Type Definitions and XSD References

D.1 XML Schema Definitions (XSD)

Common XML schema definitions according to TS 34.229-3 [28] Table G.0.1-1 are used. In addition, there are the MCX specific XML schema definitions as according to table D.1-1.

Table D.1-1: MCX specific definitions

XML Schema (XSD)	Source	Name space
RFC4354-poc-settings	RFC 4354 [52] clause 6.1	urn:oma:params:xml:ns:poc:poc-settings
RFC4661-SimpleFilter	RFC 4661 [34] clause 7	urn:ietf:params:xml:ns:simple-filter
RFC5261-patch-ops (NOTE 2)	RFC 5261 [50] clause 8	(NOTE 1)
RFC5874-xcap-diff	RFC 5874 [51] clause 4	urn:ietf:params:xml:ns:xcap-diff
IANA-resource-lists	RFC 4826 [34] clause 3.2 https://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd	urn:ietf:params:xml:ns:resource-lists
poc_listService-v1_0	OMA [37]	urn:oma:xml:poc:list-service
xdm_extensions-v1_0	OMA [39]	urn:oma:xml:xdm:extensions
xdm_rsrclst_uriusage-v1_0	OMA [40]	urn:oma:xml:xdm:resource-list:oma-uriusage
xenc-schema	W3C [41]	http://www.w3.org/2001/04/xmlenc#'
xmldsig-core-schema	W3C [42]	http://www.w3.org/2000/09/xmldsig#
TS24281_mcvideoinfo (NOTE 3)	TS 24.281 [55] Annex F.1.2	urn:3gpp:ns:mcvideoInfo:1.0
TS24281_mcvideolocationinfo	TS 24.281 [55] Annex F.3.2	urn:3gpp:ns:mcvideoLocationInfo:1.0
TS24281_mcvideoPresInfo	TS 24.281 [55] Table 8.3.1.2-1	urn:3gpp:ns:mcvideoPresInfo:1.0
TS24281_mcvideoPresInfoFA	TS 24.281 [55] Table 20.3.1.2-1	urn:3gpp:ns:mcvideoPresInfoFA:1.0
TS24281_preconfiguredRegroup	TS 24.281 [55] Annex F.6.2	urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6
TS24282_mcdatainfo	TS 24.282 [57] Annex D.1.2	"urn:3gpp:ns:mcdataInfo:1.0"
TS24282_mcdatalocationinfo	TS 24.282 [57] Annex D.4.2	urn:3gpp:ns:mcdataLocationInfo:1.0
TS24282_mcdataPresInfo	TS 24.282 [57] Table 8.4.1.2-1	"urn:3gpp:ns:mcdataPresInfo:1.0"
TS24282_mcdataPresInfoFA	TS 24.282 [57] Table 22.3.1.2-1	"urn:3gpp:ns:mcdataPresInfoFA:1.0"
TS24282_preconfiguredRegroup	TS 24.282 [57] Annex F.6.2	urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6
TS24379_mcpttaff (NOTE 4)	TS 24.379 [9] Annex F.4.2 TS 24.281 [55] Annex F.4.2 TS 24.282 [57] Annex D.3.2	urn:3gpp:ns:affiliationCommand:1.0
TS24379_mcpttinfo	TS 24.379 [9] Annex F.1.2	urn:3gpp:ns:mcpttInfo:1.0
TS24379_mcpttlocationinfo	TS 24.379 [9] Annex F.3.2	urn:3gpp:ns:mcpttLocationInfo:1.0
TS24379_mcpttsigneddoc	TS 24.379 [9] Annex F.6.2	urn:3gpp:ns:mcpttSignedDoc:1.0
TS24379_preconfiguredRegroup	TS 24.379 [9] Annex F.7.2	urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6
TS24379_mcpttPresInfo	TS 24.379 [9] Table 9.3.1.2-1	urn:3gpp:ns:mcpttPresInfo:1.0
TS24379_mcpttPresInfoFA	TS 24.379 [9] Table 9A.3.1.2-1	urn:3gpp:ns:mcpttPresInfoFA:1.0
TS24379_poc-settings (NOTE 5)	TS 24.379 [9] Table 7.4.1.2.2-2 TS 24.281 [55] Table 7.4.1.2.2-2	urn:3gpp:mcsSettings:1.0
TS24481-mcptt-group	TS 24.481 [11] clause 7.2.4.2	urn:3gpp:ns:mcpttGroupInfo:1.0
TS24481-mcpttGMOP	TS 24.481 [11] clause 7.3.3	urn:3gpp:ns:mcpttGMOP:1.0
TS24481-mcpttGKTP	TS 24.481 [11] clause 7.7.4.2	urn:3gpp:ns:mcpttGKTP:1.0
TS24484-ue-init-config	TS 24.484 [14] clause 7.2.2.3	urn:3gpp:mcptt:mcpttUEinitConfig:1.0
TS24484- mcptt-ue-config	TS 24.484 [14] clause 8.2.2.3	urn:3gpp:mcptt:mcpttUEConfig:1.0
TS24484-mcvideo-ue-config	TS 24.484 [14] clause 9.2.2.3	urn:3gpp:mcvideo:mcvideoUEConfig:1.0
TS24484-mcdata-ue-config	TS 24.484 [14] clause 10.2.2.3	urn:3gpp:mcdata:mcdataUEConfig:1.0
TS24484-mcptt-user-profile	TS 24.484 [14] clause 8.3.2.3	urn:3gpp:mcptt:user-profile:1.0
TS24484-mcvideo-user-profile	TS 24.484 [14] clause 9.3.2.3	urn:3gpp:ns:mcvideo:user-profile:1.0
TS24484-mcdata-user-profile	TS 24.484 [14] clause 10.3.2.3	urn:3gpp:ns:mcdata:user-profile:1.0
TS24484-mcptt-service-config	TS 24.484 [14] clause 8.4.2.3	urn:3gpp:ns:mcpttServiceConfig:1.0
TS24484-mcvideo-service-config	TS 24.484 [14] clause 9.4.2.3	urn:3gpp:ns:mcvideoServiceConfig:1.0

TS24484-mcdata-service-config	TS 24.484 [14] clause 10.4.2.3	urn:3gpp:ns:mcdataServiceConfig:1.0
TS33180-mcsecKMSInterface	TS 33.180 [43] Annex D.3.5.1	urn:3gpp:ns:mcsecKMSInterface:1.0
TS33180-mcsecKMSKRR	TS 33.180 [43] Annex D.4.4	urn:3gpp:ns:mcsecKMSKRR:1.0

NOTE 1: The schema does not define any target namespace but is referred to as "urn:ietf:params:xml:schema:patch-ops" e.g. in the original include statement in urn:ietf:params:xml:ns:xcap-diff.
NOTE 2: The XML schema in RFC 5261 [50] clause 8 is specified as DTD with ENTITY declarations being referred by the pattern of the schema. These references are resolved in RFC5261-patch-ops in order to get schema definitions which can be processed by TTCN-3 compilers.
NOTE 3: The schema specifies import of non-existing namespace urn:3gpp:ns:mcvideoGKTP:1.0.
Assumption: urn:3gpp:ns:mcpttGKTP:1.0 to be imported instead
NOTE 4: The schemata in TS 24.379 [9] F.4.2, TS 24.281 [55] F.4.2 and TS 24.282 [57] Annex D.1.2 are effectively the same apart from the namespace prefix ("mcpttaff" vs. "mcvideoaff" vs. "mcdataaff").
Assumption: The namespace prefix does not matter in XML documents, in general any prefix can be used.
⇒ TS24379_mcpttaff is used for MCPTT, MCVideo and MCData.
NOTE 5: The schemata in TS 24.379 [9] Table 7.4.1.2.2-2 and TS 24.281 [55] Table 7.4.1.2.2-2 are the same.
⇒ TS24379_poc-settings is used for MCPTT and MCVideo.
NOTE 6: The "XML schema for regroup using preconfigured group" in TS 24.379 [9], TS 24.281 [55] and TS 24.282 [57] have the same target namespace.

Editor's note: The XML schemas define identical types with the same name ("preconfig-group-Type", "groups-for-regroup-Type", "users-for-regroup-Type", "anyExtType") which is not allowed in one and the same namespace. As workaround these type definitions are removed from the XML schemas of TS 24.281 [55] and TS 24.282 [57].

In order to avoid ambiguities and unexpected side effects due to tool specific behaviour the schemaLocation in xs:import statements of XSD files are modified to use local references rather than references to internet locations.

The schemaLocations of imported namespaces are shown in table D.1-2. The schemaLocations of included namespaces are shown in table D.1-3.

Table D.1-2: schemaLocation of imported namespaces

XML Schema	Imported name space	Original schemaLocation	Modified schemaLocation
RFC4354-poc-settings	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
RFC4661-SimpleFilter	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
IANA-resource-lists	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
poc_listService-v1_0	urn:ietf:params:xml:ns:common-policy	http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd	../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1)
poc_listService-v1_0	urn:ietf:params:xml:ns:resource-lists	http://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd	IANA-resource-lists.xsd (NOTE 2)
xdm_extensions-v1_0	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/03/xml.xsd	../../Common/IMS_XSD/xml.xsd
xenc-schema	http://www.w3.org/2000/09/xmldsig#	xmldsig-core-schema.xsd	(no change needed)
TS24282_mcdatainfo.xsd	http://www.w3.org/2001/04/xmlenc#	http://www.w3.org/TR/xmlenc-core/xenc-schema.xsd	xenc-schema.xsd
TS24379_mcpttsigneddoc	http://www.w3.org/2000/09/xmldsig#	http://www.w3.org/TR/xmldsig-core/xmldsig-core-schema.xsd	xmldsig-core-schema.xsd
TS24484-ue-init-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484- mcptt-ue-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcvideo-ue-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcdata-ue-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcptt-user-profile	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcptt-user-profile	urn:ietf:params:xml:ns:common-policy	http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd	../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1)
TS24484-mcvideo-user-profile	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcvideo-user-profile	urn:ietf:params:xml:ns:common-policy	http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd	../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1)
TS24484-mcdata-user-profile	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
TS24484-mcdata-user-profile	urn:ietf:params:xml:ns:common-policy	http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd	../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1)
TS24484-mcptt-service-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd

XML Schema	Imported name space	Original schemaLocation	Modified schemaLocation
TS24484-mcdata-service-config	http://www.w3.org/XML/1998/namespace	http://www.w3.org/2001/xml.xsd	../../Common/IMS_XSD/xml.xsd
NOTE 1: The namespace urn:ietf:params:xml:ns:common-policy is specified in RFC 4745 and at http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd with difference in the minOccurs for identityType and validityType. Nevertheless for use in TTCN there shall be only one schema as part 9 of the TTCN-3 language [31] does not support different variants of one and the same namespace. In addition according to DIFF1 for RFC 4745 it seems that the RFC's variant is newer than the IANA's variant. NOTE 2: The namespace urn:ietf:params:xml:ns:resource-lists is specified in RFC 4826 and at http://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd with the difference of attribute anchor not being mandatory in the RFC.			

Table D.1-3: schemaLocation of included namespaces

XML Schema	Included schema	Original schemaLocation	Modified schemaLocation
RFC5874-xcap-diff	urn:ietf:params:xml:schema:patch-ops	urn:ietf:params:xml:schema:patch-ops	RFC5261-patch-ops.xsd

D.2 Common TTCN-3 Libraries

The same LibSip modules are used as according to TS 34.229-3 [28] annex G.0.2 and the same additional LibSip_MessageBodyTypes as according to TS 34.229-3 [28] annex G.1 and G.2

Annex E (normative): TTCN-3 Definitions for the MCX IPCAN test model

E.0 Introduction

The present Annex E specifies the TTCN-3 type definitions used at the system interface to configure and control the SS. The type definitions are specific for the MCX IPCAN test model.

In case of discrepancy between the content of the present Annex E and the equivalent TTCN-3 definitions / semantic requirements found in the TTCN modules provided as attachments to the present specification, the latter shall take precedence.

NOTE: This annex is automatically generated from the TTCN-3 modules provided as attachment to the present specification and containing the listed TTCN-3 type definitions.

E.1 MCX_IPCAN_ASP_TypeDefs

MCX_IPCAN_ASP_TypeDefs: Basic Type Definitions

TTCN-3 Basic Types		
MCX_IPCAN_AbsoluteCellPower_Type	integer (-150..0)	absolute cell power (dBm); same as AbsoluteCellPower_Type and NR_AbsoluteCellPower_Type
MCX_IPCAN_DeactivatedEdicatedEpsBearer_Type	MCX_IPCAN_EpsBearerIdList_Type	Deactivate dedicated EPS bearer(s) for the MCX PDN CellId : cell id of the cell Reference implementation: f_EUTRA_DeactivateEPS_BearerContext
MCX_IPCAN_RemoveQosFlow_Type	MCX_IPCAN_QosFlowList_Type	Remove QoS flow(s) from MCX PDU session CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_RemoveQosFlow

MCX_IPCAN_EpsBearerIdList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_EpsBearerIdList_Type**
Comment	
record of MCX_IPCAN_EpsBearerId_Type	

MCX_IPCAN_QosFlowList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_QosFlowList_Type**
Comment	
record of QosFlow_Identification_Type	

MCX_IPCAN_PacketFilterDirection_Type

TTCN-3 Enumerated Type	
Name	**MCX_IPCAN_PacketFilterDirection_Type**
Comment	packet filter direction according to 24.008 Table 10.5.162
downlinkOnly	
uplinkOnly	
bidirectional	

MCX_IPCAN_PacketFilterComponentPortRange_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PacketFilterComponentPortRange_Type**		
Comment			
Low	integer		
High	integer		

MCX_IPCAN_PacketFilterComponent_Type

TTCN-3 Union Type		
Name	**MCX_IPCAN_PacketFilterComponent_Type**	
Comment	24.008 Table 10.5.162	
Protocol	integer	Protocol identifier/Next header type: 17 for UDP
SingleRemotePort	integer	
RemotePortRange	MCX_IPCAN_PacketFilterComponentPortRange_Type	

MCX_IPCAN_PacketFilterComponentList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_PacketFilterComponentList_Type**
Comment	
record of MCX_IPCAN_PacketFilterComponent_Type	

MCX_IPCAN_PacketFilter_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PacketFilter_Type**		
Comment			
Direction	MCX_IPCAN_PacketFilterDirection_Type		
ComponentList	MCX_IPCAN_PacketFilterComponentList_Type		

MCX_IPCAN_PacketFilterList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_PacketFilterList_Type**
Comment	
record of MCX_IPCAN_PacketFilter_Type	

MCX_IPCAN_DedicatedEpsBearerInfo_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_DedicatedEpsBearerInfo_Type**		
Comment	Id and packet filter for dedicated EPS bearer (EUTRA)		
EpsBearerId	MCX_IPCAN_EpsBearerId_Type		EPS bearer id of the dedicated EPS bearer
PacketFilterList	MCX_IPCAN_PacketFilterList_Type		

MCX_IPCAN_QosFlowInfo_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_QosFlowInfo_Type**		
Comment	identification (PDU session id and QFI) and packet filter for dedicated Qos flow (NR5GC)		
QosFlow	QosFlow_Identification_Type		Qos flow identification
PacketFilterList	MCX_IPCAN_PacketFilterList_Type		

MCX_IPCAN_IPv4IPv6Address_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_IPv4IPv6Address_Type**		
Comment			
IPv4	charstring		
IPv6	charstring		

MCX_IPCAN_PDN_Info_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PDN_Info_Type**		
Comment			
PdnType	MCX_Registration_PDN_Type		MCX, IMS or INTERNET
EpsBearerId	MCX_IPCAN_EpsBearerId_Type		EPS bearer id of the PDN's default EPS bearer
IPAddressNW	MCX_IPCAN_IPv4IPv6Address_Type		
IPAddressUE	MCX_IPCAN_IPv4IPv6Address_Type		

MCX_IPCAN_PDN_InfoList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_PDN_InfoList_Type**
Comment	list of PDNs to be connected to during Idle Update procedure
record length (1..3) of MCX_IPCAN_PDN_Info_Type	

MCX_IPCAN_PduSession_Info_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PduSession_Info_Type**		
Comment			
PdnType	MCX_Registration_PDN_Type		MCX, IMS or INTERNET
APN	charstring		
IPAddressNW	MCX_IPCAN_IPv4IPv6Address_Type		
IPAddressUE	MCX_IPCAN_IPv4IPv6Address_Type		

MCX_IPCAN_PduSession_InfoList_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_PduSession_InfoList_Type**
Comment	list of PDU sessions to be connected to during Idle Update procedure
record length (0..3) of MCX_IPCAN_PduSession_Info_Type	

MCX_IPCAN_ASP_CommonReqPart_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_ASP_CommonReqPart_Type**		
Comment	Common Part of system requests		
CellId	AnyCellId_Type	opt	not present in case of InitRequest and ReleaseRequest, mandatory otherwise
CnfFlag	boolean		true: SS shall send a confirmation when the request has been executed false: SS shall not send a confirmation for this request

MCX_IPCAN_ASP_CommonIndAckPart_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_ASP_CommonIndAckPart_Type**		
Comment	Common Part of system indications and acknowledgements		
CellId	AnyCellId_Type		

MCX_IPCAN_Init_Type

TTCN-3 Record Type	
Name	**MCX_IPCAN_Init_Type**
Comment	Initialisation of the system simulator. CellId : not present Reference implementation 4G: f_EUTRA_Init with p_SysinfoCombination:=c1, p_HandleULDataUM:=HANDLE_UM_DATA, p_UseBigGrants:=USE_BIG_GRANT Reference implementation 5G: f_NR5GC_Init

MCX_IPCAN_CellConfig_Type

TTCN-3 Record Type	
Name	**MCX_IPCAN_CellConfig_Type**
Comment	Create a cell and send out system information. CellId : cell id of the cell to be created Reference implementation 4G: f_EUTRA_CellConfig_Def Reference implementation 5G: f_NR_CellConfig_Def

MCX_IPCAN_CellPower_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_CellPower_Type**		
Comment	Set the cell power of a cell. CellId : cell id of the cell Reference implementation 4G: f_EUTRA_SetCellPower Reference implementation 5G: f_NR_SetCellPower		
AbsoluteCellPower	MCX_IPCAN_AbsoluteCellPower_Type		

MCX_IPCAN_InitialRegistration4G_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_InitialRegistration4G_Type**		
Comment	Perform initial registration procedure according to 37.579-1 clause 5.4.1A and Table 5.4.2.2.3-1 steps 2 - 16A to bring the UE into state 3 according to Figure 4.5.1-1 in 36.508. As described in clause 5.4.1A of 37.579-1 after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. NOTE: The UE does not need to register to MCX during initial registration after switch-on. CellId : cell id of the cell Reference implementation: f_MCX_EUTRA_IPCAN_IdleUpdated		
PdnList	MCX_IPCAN_PDN_InfoList_Type		list of PDNs the UE is expected to connect to during Idle Update procedure; the list may or may not contain MCX
DefaultAPN	octetstring		APN to be used for the PDN (if any) for which the UE does not provide an APN; APN encoded according to 23.003 clause 9.1 and RFC 1035 clause 4.1.2

MCX_IPCAN_InitialRegistration5G_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_InitialRegistration5G_Type**		
Comment	Perform initial registration procedure according to 37.579-1 clause 5.4.1B and Table 5.4.2.3.3-1 steps 1 - 15 to bring the UE into state 1N-A according to 38.508-1 clause 4.4A. As described in clause 5.4.1B of 37.579-1 after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. NOTE: The UE does not need to register to MCX during initial registration after switch-on. CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_InitialRegistration (f_NR5GC_Registration, a_NR5GC_AdditionalPDUSessionEstablishmentReq, f_MCX_NR5GC_IPCAN_AddPduSession)		
PduSessionList	MCX_IPCAN_PduSession_InfoList_Type		list of PDU sessions the UE is expected to connect to during Idle Update procedure; the list may or may not contain MCX and it may be empty

MCX_IPCAN_ExplicitMCXRegistration4G_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_ExplicitMCXRegistration4G_Type**		
Comment	Procedure used, if the UE is configured not to connect automatically to the MCX PDN after switch-on: In this case after initial registration the UE gets triggered from TTCN by an MMI command to start MCX registration and this procedure is started to perform steps 19 - 26 of 37.579-1 Table 5.4.2.2.3-1. CellId : cell id of the cell Reference implementation: f_MCX_EUTRA_IPCAN_ConnectToAdditionalPDN		
PdnInfo	MCX_IPCAN_PDN_Info_Type		

MCX_IPCAN_ExplicitMCXRegistration5G_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_ExplicitMCXRegistration5G_Type**		
Comment	Procedure used, if the UE is configured not to connect automatically to the MCX PDN after switch-on: In this case after initial registration the UE gets triggered from TTCN by an MMI command to start MCX registration and this procedure is started to perform steps 1 - 9 of 37.579-1 Table 5.4.2.3.3-2. CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_AddPduSession		
PduSessionInfo	MCX_IPCAN_PduSession_Info_Type		

MCX_IPCAN_RRCConnectionSetup4G_MO_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_RRCConnectionSetup4G_MO_Type**		
Comment	MO RRC connection establishment according to 37.579-1 table 5.4.3.2.3-1 CellId : cell id of the cell Reference implementation: f_MCX_EUTRA_IPCAN_RRCConnectionEstablishment_MO		
DedicatedEpsBearers	MCX_IPCAN_EpsBearerIdList_Type		In case of a pre-established session: list of dedicated EPS bearers for which the DRBs need to be established; otherwise {}

MCX_IPCAN_RRCConnectionSetup5G_MO_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_RRCConnectionSetup5G_MO_Type**		
Comment	MO RRC establishment according to 37.579-1 table 5.4.3.3.3-1 CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_RRCEstablishment_MO		
QosFlows	MCX_IPCAN_QosFlowList_Type		In case of a pre-established session: list of QoS flows for which the DRBs need to be established; otherwise {}

MCX_IPCAN_RRCConnectionSetup4G_MT_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_RRCConnectionSetup4G_MT_Type**		
Comment	MT RRC connection establishment according to 37.579-1 table 5.4.4.2.3-1 CellId : cell id of the cell Reference implementation: f_MCX_EUTRA_IPCAN_RRCConnectionEstablishment_MT		
DedicatedEpsBearers	MCX_IPCAN_EpsBearerIdList_Type		In case of a pre-established session: list of dedicated EPS bearers for which the DRBs need to be established; otherwise {}

MCX_IPCAN_RRCConnectionSetup5G_MT_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_RRCConnectionSetup5G_MT_Type**		
Comment	MT RRC establishment according to 37.579-1 table 5.4.4.3.3-1 CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_RRCEstablishment_MT		
QosFlows	MCX_IPCAN_QosFlowList_Type		In case of a pre-established session: list of QoS flows for which the DRBs need to be established; otherwise {}

MCX_IPCAN_ActivateDedicatedEpsBearer_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_ActivateDedicatedEpsBearer_Type**
Comment	Establish dedicated EPS bearer(s) for the MCX PDN CellId : cell id of the cell Reference implementation: f_MCX_EUTRA_IPCAN_ActivateDedicatedEpsBearer
record of MCX_IPCAN_DedicatedEpsBearerInfo_Type	

MCX_IPCAN_AddQosFlow_Type

TTCN-3 Record of Type	
Name	**MCX_IPCAN_AddQosFlow_Type**
Comment	Add QoS flow(s) to MCX PDU session CellId : cell id of the cell Reference implementation: f_MCX_NR5GC_IPCAN_AddQosFlow
record of MCX_IPCAN_QosFlowInfo_Type	

MCX_IPCAN_RRCConnectionRelease_Type

TTCN-3 Record Type	
Name	**MCX_IPCAN_RRCConnectionRelease_Type**
Comment	Relase RRC connection CellId : cell id of the cell Reference implementation 4G: f_EUTRA_RRC_ConnectionRelease Reference implementation 5G: f_NR_RRCRelease

MCX_IPCAN_Release_Type

TTCN-3 Record Type	
Name	**MCX_IPCAN_Release_Type**
Comment	Release all configuration at the system simulator; CellId : not present Reference implementation 4G: f_EUTRA_Postamble with p_E_State:=E1_IDLE Reference implementation 5G: f_NR_Postamble with p_UE_State:=STATE_IDLE_1A

MCX_IPCAN_SystemRequest_Type

TTCN-3 Union Type		
Name	**MCX_IPCAN_SystemRequest_Type**	
Comment		
Init	MCX_IPCAN_Init_Type	
CellConfig	MCX_IPCAN_CellConfig_Type	
CellPower	MCX_IPCAN_CellPower_Type	
InitialRegistration4G	MCX_IPCAN_InitialRegistration4G_Type	
InitialRegistration5G	MCX_IPCAN_InitialRegistration5G_Type	
ExplicitMCXRegistration4G	MCX_IPCAN_ExplicitMCXRegistration4G_Type	
ExplicitMCXRegistration5G	MCX_IPCAN_ExplicitMCXRegistration5G_Type	
RRCConnectionSetup4G_MO	MCX_IPCAN_RRCConnectionSetup4G_MO_Type	
RRCConnectionSetup5G_MO	MCX_IPCAN_RRCConnectionSetup5G_MO_Type	
RRCConnectionSetup4G_MT	MCX_IPCAN_RRCConnectionSetup4G_MT_Type	
RRCConnectionSetup5G_MT	MCX_IPCAN_RRCConnectionSetup5G_MT_Type	
ActivateDedicatedEpsBearer	MCX_IPCAN_ActivateDedicatedEpsBearer_Type	
AddQosFlow	MCX_IPCAN_AddQosFlow_Type	
DeactivateDedicatedEpsBearer	MCX_IPCAN_DeactivateDedicatedEpsBearer_Type	
RemoveQosFlow	MCX_IPCAN_RemoveQosFlow_Type	
RRCConnectionRelease	MCX_IPCAN_RRCConnectionRelease_Type	
Release	MCX_IPCAN_Release_Type	

MCX_IPCAN_SYSTEM_CTRL_REQ

TTCN-3 Record Type			
Name	**MCX_IPCAN_SYSTEM_CTRL_REQ**		
Comment			
Common	MCX_IPCAN_ASP_CommonReqPart_Type		
Request	MCX_IPCAN_SystemRequest_Type		

MCX_IPCAN_InitConfirmation_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_InitConfirmation_Type**		
Comment			
RanType	IPCAN_RAN_Type		EUTRA_FDD or EUTRA_TDD for EUTRA, NR for NR5GC

MCX_IPCAN_SystemConfirmation_Type

TTCN-3 Union Type		
Name	**MCX_IPCAN_SystemConfirmation_Type**	
Comment		
Init	MCX_IPCAN_InitConfirmation_Type	Confirmation for Init request: carries the RAN Type
Other	Null_Type	Confirmation for any other request
Error	charstring	Error indication when the corresponding request causes error; the charstring may be used to provide information about the error

MCX_IPCAN_SYSTEM_CTRL_CNF

TTCN-3 Record Type			
Name	**MCX_IPCAN_SYSTEM_CTRL_CNF**		
Comment			
Confirmation	MCX_IPCAN_SystemConfirmation_Type		

MCX_IPCAN_PduSessionEstReqInd_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PduSessionEstReqInd_Type**		
Comment			
PduSessionId	integer		

MCX_IPCAN_SystemIndication_Type

TTCN-3 Union Type		
Name	**MCX_IPCAN_SystemIndication_Type**	
Comment		
PduSessionEstReqInd	MCX_IPCAN_PduSessionEstReqInd_Type	

MCX_IPCAN_SYSTEM_IND

TTCN-3 Record Type			
Name	**MCX_IPCAN_SYSTEM_IND**		
Comment			
Common	MCX_IPCAN_ASP_CommonIndAckPart_Type		
Indication	MCX_IPCAN_SystemIndication_Type		

MCX_IPCAN_PduSessionEstReqAck_Type

TTCN-3 Record Type			
Name	**MCX_IPCAN_PduSessionEstReqAck_Type**		
Comment			
QFI	integer		

MCX_IPCAN_SystemAcknowledgement_Type

TTCN-3 Union Type		
Name	**MCX_IPCAN_SystemAcknowledgement_Type**	
Comment		
PduSessionEstReqAck	MCX_IPCAN_PduSessionEstReqAck_Type	

MCX_IPCAN_SYSTEM_ACK

TTCN-3 Record Type			
Name	**MCX_IPCAN_SYSTEM_ACK**		
Comment			
Common	MCX_IPCAN_ASP_CommonIndAckPart_Type		
Acknowledgement	MCX_IPCAN_SystemAcknowledgement_Type		

MCX_IPCAN_SYSTEM_CTRL_PORT

TTCN-3 Port Type		
Name	**MCX_IPCAN_SYSTEM_CTRL_PORT**	
Comment		
out	MCX_IPCAN_SYSTEM_CTRL_REQ	
in	MCX_IPCAN_SYSTEM_CTRL_CNF	

MCX_IPCAN_SYSTEM_IND_PORT

TTCN-3 Port Type		
Name	**MCX_IPCAN_SYSTEM_IND_PORT**	
Comment		
out	MCX_IPCAN_SYSTEM_ACK	
in	MCX_IPCAN_SYSTEM_IND	

E.2 MCX_CommonIPCAN

MCX_CommonIPCAN: Basic Type Definitions

TTCN-3 Basic Types		
MCX_IPCAN_EpsBearerId_Type	integer (5..13)	EPS bearer id: the associated DRB Id shall always be EpsBearerId - 4

MCX_Registration_PDN_Type

TTCN-3 Enumerated Type	
Name	**MCX_Registration_PDN_Type**
Comment	Type definition for PIXITs px_MCX_InitialRegistration_TypeOf1stPDN, px_MCX_InitialRegistration_TypeOf2ndPDN, px_MCX_InitialRegistration_TypeOf3rdPDN; (see 37.579-1 clause 5.4.1A) NOTE: 'ims', 'internet', 'mcx' and 'none' cannot be used as 'none' is a keyword; 'IMS', 'INTERNET', 'MCX' and 'NONE' cannot be used as 'IMS' is a port name of various PTCs
imsPDN	
internetPDN	
mcxPDN	
noPDN	

E.3 CommonDefs

CommonDefs: Basic Type Definitions

TTCN-3 Basic Types		
Null_Type	boolean (true)	dummy type for 'typeless' fields in unions

NR_CellId_Type

TTCN-3 Enumerated Type	
Name	**NR_CellId_Type**
Comment	
nr_Cell_NonSpecific	
nr_Cell1	
nr_Cell2	
nr_Cell3	
nr_Cell4	
nr_Cell6	
nr_Cell10	
nr_Cell11	
nr_Cell12	
nr_Cell13	
nr_Cell14	
nr_Cell23	
nr_Cell28	
nr_Cell29	
nr_Cell30	
nr_Cell31	

EUTRA_CellId_Type

TTCN-3 Enumerated Type	
Name	**EUTRA_CellId_Type**
Comment	
eutra_Cell_NonSpecific	
eutra_Cell1	
eutra_Cell2	
eutra_Cell3	
eutra_Cell4	
eutra_Cell6	
eutra_Cell10	
eutra_Cell11	
eutra_Cell12	
eutra_Cell13	
eutra_Cell14	
eutra_Cell23	
eutra_Cell28	
eutra_Cell29	
eutra_Cell30	
eutra_Cell31	
eutra_CellA	
eutra_CellB	
eutra_CellC	
eutra_CellD	
eutra_CellE	
eutra_CellG	
eutra_CellH	
eutra_CellI	
eutra_CellJ	
eutra_CellK	
eutra_CellL	
eutra_CellM	

AnyCellId_Type

TTCN-3 Union Type		
Name	**AnyCellId_Type**	
Comment		
Eutra	EUTRA_CellId_Type	
Nr	NR_CellId_Type	

QosFlow_Identification_Type

TTCN-3 Record Type			
Name	**QosFlow_Identification_Type**		
Comment			
PDU_SessionId	integer		TS 24.007 clause 11.2.3.1b
QFI	integer		TS 24.501 Table 11.2.3.1c.1

IPCAN_RAN_Type

TTCN-3 Enumerated Type	
Name	**IPCAN_RAN_Type**
Comment	radio access network technology used by the IPCAN PTC
GERAN	
UTRAN_FDD	
UTRAN_TDD	
EUTRA_FDD	
EUTRA_TDD	
WLAN	
NR	

E.4 References to TTCN-3

References to TTCN-3		
MCX_IPCAN_ASP_TypeDefs	IPCAN_IpcanModel/MCX_IPCAN_ASP_TypeDefs.ttcn	Rev 41472
MCX_CommonIPCAN	IPCAN_Common/MCX_CommonIPCAN.ttcn	Rev 41607
CommonDefs	Common/CommonDefs.ttcn	Rev 41588

Annex F (informative): Procedures of the MCX IPCAN test model

F.1 Introduction

This annex describes the major procedures for the SS based E-UTRA/EPC or NR/5GC implementation of the MCX IPCAN test model. The procedures are described in tables showing the primitives exchanged between TTCN and the system simulator at the IPCAN model specific ports SYS_REQ and SYS_IND as shown in Figure 5.2.1-1 and the related actions.

In general, the procedures correspond to the procedures specified in TS 37.579-1 [2] clause 5.4.

F.2 Procedures

F.2.1 Initialisation

This procedure is used to initialise the system simulator and to retrieve the RAN type (EUTRA_FDD, EUTRA_TDD, NR) to be used.

Table F.2.1-1: Initialisation

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
1	Common initialisation	Init (NOTE 1)	-->	-
2	Confirmation that initialisation is completed	Confirmation (Init) (NOTE 2)	<--	RAN type
3	Cell configuration	CellConfig (NOTE 1)	-->	cell id
4	Confirmation that cell configuration is completed	Confirmation (Other) (NOTE 2)	<--	-
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).				

F.2.2 Initial Registration

This procedure is used for initial registration and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.2.

Table F.2.2-1: Initial Registration

Step	Action/Step	System Primitives at SYS_REQ/IND port	Direction TTCN-SS	Relevant parameters / Comments
1	Configuration of the cell power	CellPower (NOTE 1)	-->	cell id, absolute cell power
-	EXCEPTION: IF the RAN type is EUTRA THEN step 2a1 is perfomed	-	-	-
2a1	Start of IP handling (Address, IMS, HTTP servers) for MCX	-	-	cell id, MCX default EPS bearer id
3	Switch on the UE	-	-	-
4	Start of initial registration	EUTRA: InitialRegistration4G (NOTE 1)	-->	list of PDN infos: PDN type, EPS bearer id, IP addresses
		NR5GC: InitialRegistration5G (NOTE 1)	-->	list of PDU session infos: PDN type, APN, IP addresses
-	EXCEPTION: Depending on the UE configuration, IF the UE does not register to MCX automatically at initial registration THEN steps 5a1-5a5 are performed	-	-	-
5a1	Confirmation that the initial registration is completed	Confirmation (Other) (NOTE 2)	<--	(NOTE 5)
5a2	Request of RRC connection release	RRCConnectionRelease (NOTE 1)	-->	-
5a3	Confirmation for RRC connection release	Confirmation (Other) (NOTE 2)	<--	
5a4	Make the UE (MCX client) request service authorisation/configuration	-	-	-
5a5	Start of explicit MCX registration	EUTRA: ExplicitMCXRegistration4G (NOTE 1)	-->	MCX PDN info: PDN type, EPS bearer id, IP addresses
		NR5GC: ExplicitMCXRegistration5G (NOTE 1)	-->	MCX PDU session info: PDN type, APN, IP addresses
-	EXCEPTION: IF the RAN type is NR/5GC THEN Steps 6a1-6a3 are performed (NOTE 6)	-	-	-
6a1	Indication of PDU SESSION ESTABLISHMENT REQUEST	PduSessionEstReqInd (NOTE 3)	<--	PDU session id
6a2	Start of IP handling (Address, IMS, HTTP servers) for MCX	-	-	cell id, MCX default QoS flow (PDU session id, QFI)
6a3	Acknowledgement of PDU session id and assignment of the QFI to be used for MCX default QoS flow	PduSessionEstReqAck (NOTE 4)	-->	QFI
7	HTTP and SIP signalling according to TS 37.579-1 [2] Table 5.3.2.2.1-1	-	-	-
8	Request of RRC connection release	RRCConnectionRelease (NOTE 1)	-->	-
9	Confirmation for RRC connection release	Confirmation (Other) (NOTE 2)	<--	-

NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).
NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).
NOTE 3: MCX_IPCAN_SYSTEM_IND with Indication as specified in annex E (MCX_IPCAN_SystemIndication_Type).
NOTE 4: MCX_IPCAN_SYSTEM_ACK with Acknowledgement as specified in annex E (MCX_IPCAN_SystemAcknowledgement_Type)
NOTE 5: A confirmation is requested by the InitialRegistration at step 4 if the UE does not register to MCX automatically.
NOTE 6: Basically, the SS should indicate the PDU session id provided by the UE in the PDU SESSION ESTABLISHMENT REQUEST (step 6a1) and wait for the acknowledgement before sending the PDU SESSION ESTABLISHMENT ACCEPT. Nevertheless, it is up to SS implementation how to know the PDU session id, but the SS is expected to indicate the PDU session id early enough so that the HTTP server is in place when the initial HTTP request from the UE needs to be handled in TTCN.

F.2.3 MO RRC connection setup

This procedure is used for MO RRC connection setup and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.3: Steps 2-8 of Table 5.4.3.2.3-1 for EUTRA and steps 1-7 of Table 5.4.3.3.3-1 for NR5GC, with m=1 at step 7 of Table 5.4.3.2.3-1 and at step 6 of Table 5.4.3.3.3-1 in case of a pre-established session and m=0 otherwise.

Table F.2.3-1: MO RRC connection setup

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
1	Request of MO RRC connection setup	EUTRA: RRCConnectionSetup4G_MO (NOTE 1)	-->	list of dedicated EPS bearer ids (NOTE 2)
		NR5GC: RRCConnectionSetup5G_MO (NOTE 1)	-->	list of dedicated QoS flows (PDU session id, QFI) (NOTE 2)
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: In case of pre-established session, the RRC connection setup request provides a list of dedicated EPS bearer ids (EUTRA/EPC) or dedicated QoS flows (NR/5GC) with a single entry. When there is no pre-established session, the list is empty.				

F.2.4 MT RRC connection setup

This procedure is used for MT RRC connection setup and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.4: Steps 1-8 of Table 5.4.4.2.3-1 for EUTRA and steps 1-8 of Table 5.4.4.3.3-1 for NR5GC, with m=1 at step 7 of Table 5.4.4.2.3-1 and at step 7 of Table 5.4.4.3.3-1 in case of a pre-established session and m=0 otherwise.

Table F.2.4-1: MT RRC connection setup

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
1	Request of MT RRC connection setup	EUTRA: RRCConnectionSetup4G_MT (NOTE 1)	-->	list of dedicated EPS bearer ids (NOTE 2)
		NR5GC: RRCConnectionSetup5G_MT (NOTE 1)	-->	list of dedicated QoS flows (PDU session id, QFI) (NOTE 2)
2	Confirmation for RRC connection establishment	Confirmation (Other) (NOTE 3)	<--	-
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: In case of pre-established session, the RRC connection setup request provides a list of dedicated EPS bearer ids (EUTRA/EPC) or dedicated QoS flows (NR/5GC) with a single entry. When there is no pre-established session, the list is empty. NOTE 3: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).				

F.2.5 MO communication establishment

This procedure is used for MO communication establishment and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.3: Steps 2-16a3 of Table 5.4.3.2.3-1 for EUTRA and steps 1-8a3 of Table 5.4.3.3.3-1 for NR5GC, with m=0 at step 7 of Table 5.4.3.2.3-1 and at step 6 of Table 5.4.3.3.3-1.

Table F.2.5-1: MO communication establishment

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
1	MO RRC connection setup as specified in Table F.2.3-1	-	-	-
2	SIP signalling: MO INVITE	-	-	-
-	EXCEPTION: In parallel to steps 3a1-5 the SIP signalling for completion of the MO call establishment happens (NOTE 3)	-	-	-
-	EXCEPTION: Steps 3a1-3b1 depend on the kind of MCX service.	-	-	-
3a1	IF the MCX service is MCPTT or MCVideo THEN RTP/RTCP loopback gets configured			cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC)
3b1	ELSE IF the MCX service is MCData THEN MSRP server gets started in TTCN	-	-	cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC)
4	Activate/Add dedicated resources	EUTRA: ActivateDedicatedEpsBearer (NOTE 1)	-->	dedicated EPS bearer id, packet filters
		NR5GC: AddQosFlow (NOTE 1)	-->	dedicated QoS flow (PDU session id, QFI), packet filters
5	Confirmation for establishment of dedicated resources	Confirmation (Other) (NOTE 2)	<--	-
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type). NOTE 3: The UE is expected to send the SIP ACK not before dedicated resources are established.				

F.2.6 MT communication establishment

This procedure is used for MT communication establishment and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.4: Steps 1-17a3 of Table 5.4.4.2.3-1 for EUTRA and steps 1-9a3 of Table 5.4.4.3.3-1 for NR5GC, with m=0 at step 7 of Table 5.4.4.2.3-1 and at step 7 of Table 5.4.4.3.3-1.

Table F.2.6-1: MT communication establishment

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
1	MT RRC connection setup as specified in Table F.2.4-1	-	-	-
-	EXCEPTION: In parallel to steps 3a1-4 the SIP signalling for MT call establishment happens	-	-	-
-	EXCEPTION: Steps 3a1-3b1 depend on the kind of MCX service.	-	-	-
3a1	IF the MCX service is MCPTT or MCVideo THEN RTP/RTCP loopback gets configured			cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC)
3b1	ELSE IF the MCX service is MCData THEN MSRP server gets started in TTCN	-	-	cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC)
4	Activate/Add dedicated resources	EUTRA: ActivateDedicatedEpsBearer (NOTE 1)	-->	dedicated EPS bearer id, packet filters
		NR5GC: AddQosFlow (NOTE 1)	-->	dedicated QoS flow (PDU session id, QFI), packet filters
5	Confirmation for establishment of dedicated resources	Confirmation (Other) (NOTE 2)	<--	-
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).				

F.2.7 Connection release

This procedure is used for communication release and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.14.

Table F.2.7-1: Connection release

Step	Action/Step	System Primitives at SYS_REQ port	Direction TTCN-SS	Relevant parameters / Comments
-	EXCEPTION: IF there is a dedicated bearer or QoS flow AND no pre-established session is established OR a pre-established session shall be released THEN steps 1a1-1a2 are performed.	-	-	-
1a1	Deactivate/Remove dedicated resources	EUTRA: DeactivateDedicatedEpsBearer (NOTE 1)	-->	dedicated EPS bearer id
		NR5GC: RemoveQosFlow (NOTE 1)	-->	dedicated QoS flow (PDU session id, QFI
1a2	Confirmation for removal of dedicated resources	Confirmation (Other) (NOTE 2)	<--	-
2	Request of RRC connection release	RRCConnectionRelease (NOTE 1)	-->	-
3	Confirmation for RRC connection release	Confirmation (Other) (NOTE 2)	<--	-
NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type). NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).				

Annex G (informative): Change history

Change history of TS 36.579-5 up to v17.7.0							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-02	RAN5#74	R5-171302	-	-	-	Introduction of TS 36.579-5.	0.0.1
2018-03	RAN5#78	R5-180687	-	-	-	Implements changes agreed in R5-180618 "MCPTT: Initial Test Model" R5-180657 "Various updates to 36579-5"	0.1.0
2018-03	RAN#79	RP-180130	-	-	-	Draft version for information purposes to the RAN Plenary	1.0.0
2018-05	RAN5#79	R5-182437	-	-	-	Implements changes agreed in R5-183163 R5-183164	2.0.0
2018-06	RAN#80	RP-180655	-	-	-	put under revision control as v13.0.0 with small editorial changes	13.0.0
2018-09	RAN#81	R5-184081	0001	-	F	MCPTT: Test Model updates	13.1.0
2018-12	RAN#82	R5-192380	0002	1	F	Routine maintenance for TS 36.579-5	13.2.0
2019-06	RAN#84	R5-195221	0003	1	F	Routine maintenance for TS 36.579-5	13.3.0
2019-06	RAN#84	R5-195222	0004	1	F	Introduction of MCPTT test model over IP	13.3.0
2019-12	RAN#86	R5-199050	0005	1	F	Routine maintenance for TS 36.579-5	13.4.0
2020-03	RAN#87	R5-201152	0006	1	F	Routine maintenance for TS 36.579-5	13.5.0
2020-06	RAN#88	R5-203077	0007	1	F	Routine maintenance for TS 36.579-5	13.6.0
2020-09	RAN#89	R5-203554	0008	-	F	Routine maintenance for TS 36.579-5	13.7.0
2020-09	RAN#89	R5-204377	0010	-	F	Updates to introduce support for MCVideo and MCData	14.0.0
2020-12	RAN#90	R5-206465	0011	1	F	Routine maintenance for TS 36.579-5	14.1.0
2021-03	RAN#91	R5-210255	0012	-	F	Routine maintenance for TS 36.579-5	14.2.0
2021-06	RAN#92	R5-213670	0017	1	F	Routine maintenance for TS 36.579-5	14.3.0
2021-09	RAN#93	R5-216343	0019	1	F	Routine maintenance for TS 36.579-5	14.4.0
2021-09	RAN#93	R5s210779	0018	-	B	Addition of MCPTT test case 6.1.1.2 to the MCX ATS	14.4.0
2021-09	RAN#93	R5s210897	0013	1	B	Addition of MCPTT test case 5.1 to the MCX ATS	14.4.0
2021-09	RAN#93	R5s210899	0014	1	B	Addition of MCPTT test case 6.1.1.1 to the MCX ATS	14.4.0
2021-09	RAN#93	R5s211178	0020	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	14.4.0
2021-12	RAN#94	R5-216753	0060	-	F	Routine maintenance for TS 36.579-5	14.5.0
2021-12	RAN#94	R5s211359	0046	-	B	Addition of MCPTT test case 6.2.1 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211361	0047	-	B	Addition of MCPTT test case 6.2.2 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211363	0048	-	B	Addition of MCPTT test case 6.2.3 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211365	0049	-	B	Addition of MCPTT test case 6.2.4 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211367	0050	-	B	Addition of MCPTT test case 6.2.5 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211369	0051	-	B	Addition of MCPTT test case 6.2.6 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211371	0052	-	B	Addition of MCPTT test case 6.2.7 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211373	0053	-	B	Addition of MCPTT test case 6.2.8 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211377	0055	-	B	Addition of MCPTT test case 6.2.15 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211381	0057	-	B	Addition of MCPTT test case 6.2.17 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211463	0021	1	B	Addition of MCPTT test case 5.3 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211469	0059	-	F	Configuration of SRTP	14.5.0
2021-12	RAN#94	R5s211617	0023	1	B	Addition of MCPTT test case 6.1.1.4 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211618	0024	1	B	Addition of MCPTT test case 6.1.1.8 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211619	0025	1	B	Addition of MCPTT test case 6.1.1.9 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211620	0026	1	B	Addition of MCPTT test case 6.1.1.11 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211621	0027	1	B	Addition of MCPTT test case 6.1.1.12 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211628	0028	1	B	Addition of MCPTT test case 6.1.1.13 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211629	0029	1	B	Addition of MCPTT test case 6.1.1.14 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211630	0030	1	B	Addition of MCPTT test case 6.1.2.7 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211631	0031	1	B	Addition of MCPTT test case 6.1.2.8 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211632	0032	1	B	Addition of MCPTT test case 6.1.2.9 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211633	0033	1	B	Addition of MCPTT test case 6.1.2.10 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211634	0034	1	B	Addition of MCPTT test case 6.1.2.11 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211635	0035	1	B	Addition of MCPTT test case 6.1.2.12 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211688	0022	1	B	Addition of MCPTT test case 6.1.1.3 to the MCX ATS	14.5.0
2021-12	RAN#94	R5s211695	0061	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	14.5.0
2022-03	RAN#95	R5-220488	0070	-	F	Routine maintenance for TS 36.579-5	14.6.0
2022-03	RAN#95	R5s220129	0062	-	B	Addition of MCPTT test case 5.4 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220131	0063	-	B	Addition of MCPTT test case 6.1.1.5 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220133	0064	-	B	Addition of MCPTT test case 6.1.1.6 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220135	0065	-	B	Addition of MCPTT test case 6.1.1.10 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220137	0066	-	B	Addition of MCPTT test case 6.1.1.16 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220139	0067	-	B	Addition of MCPTT test case 6.1.1.18 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220141	0068	-	B	Addition of MCPTT test case 6.1.1.21 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220143	0069	-	B	Addition of MCPTT test case 6.2.16 to the MCX ATS	14.6.0
2022-03	RAN#95	R5s220344	0071	-	F	Correction of MCPTT ambient listening test cases	14.6.0
2022-03	RAN#95	R5s220346	0072	-	F	Update of function f_URL_Decoding	14.6.0

2022-03	RAN#95	R5s220437	0073	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	14.6.0
2022-06	RAN#96	R5s220559	0074	-	F	Correction of f_MCX_InviteMT_SendINVITE	14.7.0
2022-09	RAN#97	R5-223960	0076	-	F	Routine maintenance for TS 36.579-5	14.8.0
2022-09	RAN#97	R5s220869	0075	-	F	MCX Implementation Issues	14.8.0
2022-09	RAN#97	R5s220992	0077	-	B	Addition of MCPTT test case 5.6 to the MCX ATS	14.8.0
2022-09	RAN#97	R5s220994	0078	-	B	Addition of MCPTT test case 6.1.1.17 to the MCX ATS	14.8.0
2022-09	RAN#97	R5s220996	0079	-	B	Addition of MCPTT test case 6.1.1.20 to the MCX ATS	14.8.0
2022-09	RAN#97	R5s221072	0086	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	15.0.0
2022-12	RAN#98	R5s221283	0082	1	B	Addition of MCData test case 5.1 to the MCX ATS	15.1.0
2022-12	RAN#98	R5s221284	0083	1	B	Addition of MCData test case 6.1.1 to the MCX ATS	15.1.0
2022-12	RAN#98	R5s221285	0084	1	B	Addition of MCData test case 6.1.3 to the MCX ATS	15.1.0
2022-12	RAN#98	R5s221286	0085	1	B	Addition of MCData test case 6.1.4 to the MCX ATS	15.1.0
2022-12	RAN#98	R5s221361	0091	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	15.1.0
2022-12	RAN#98	-	-	-	-	upgrade to Rel16 with no change	16.0.0
2022-12	RAN#98	R5s221241	0088	-	F	Rel-17 Sep'22 baseline upgrade for MCX Test Suites	17.0.0
2023-03	RAN#99	R5-230148	0092	-	F	Routine maintenance for TS 36.579-5	17.1.0
2023-03	RAN#99	R5s230192	0090	1	F	Correction of function fl_MCPTT_Info_AmbientListening_RX	17.1.0
2023-06	RAN#100	R5-232221	0093	-	F	Routine maintenance for TS 36.579-5	17.2.0
2023-06	RAN#100	R5-232604	0094	-	F	Update to MCX PIXITs for MCData Functional Alias	17.2.0
2023-09	RAN#101	R5-233868	0099	-	F	Routine maintenance for TS 36.579-5	17.3.0
2023-09	RAN#101	R5s230578	0095	1	B	Addition of MCPTT test case 5.8 to the MCX ATS	17.3.0
2023-09	RAN#101	R5s230579	0096	1	B	Addition of MCPTT test case 6.1.1.19 to the MCX ATS	17.3.0
2023-09	RAN#101	R5s230580	0097	1	B	Addition of MCPTT test case 6.1.2.13 to the MCX ATS	17.3.0
2023-09	RAN#101	R5s230581	0098	1	B	Addition of MCPTT test case 6.2.14 to the MCX ATS	17.3.0
2023-09	RAN#101	R5s230672	0100	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A	17.3.0
2023-12	RAN#102	R5-237446	0102	1	F	Routine maintenance for TS 36.579-5	17.4.0
2024-03	RAN#103	R5-241686	0105	1	F	Addition of new PIXITs for REL-16 TCs	17.5.0
2024-03	RAN#103	R5s240111	0103	-	F	Correction of MMI in test cases 6.1.1.19 and 6.1.2.13	17.5.0
2024-03	RAN#103	R5s240148	0104	-	F	Correction of f_MCX_IMS_ServiceAuthorization	17.5.0
2024-03	RAN#103	R5s240241	0107	-	F	Update of TTCN type definitions, Annex E	17.5.0
2024-06	RAN#104	R5-242399	0111	-	F	Routine maintenance for TS 36.579-5	17.6.0
2024-06	RAN#104	R5s240279	0108	-	F	Corrections for MCX registration	17.6.0
2024-06	RAN#104	R5s240332	0106	1	F	Misc corrections for MCX	17.6.0
2024-06	RAN#104	R5s240360	0112	-	F	Correction CSK upload	17.6.0
2024-06	RAN#104	R5s240420	0114	-	F	Update of TTCN type definitions, Annex E	17.6.0
2024-09	RAN#105	R5-245598	0120	-	F	Routine maintenance for TS 36.579-5	17.7.0
2024-09	RAN#105	R5-245674	0119	1	F	Addition of MCData Message Store Function terminology	17.7.0
2024-09	RAN#105	R5s240492	0115	1	F	Correction to the request URI template	17.7.0

NOTE: The table above will not be further updated in the future. It shows all TS 36.579-5 CRs taken over into TS 37.579-5 v0.0.1.

Change history of TS 37.579-5							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-11	RAN5#105	R5-246402	-	-	-	TS 36.579-5 v17.7.0 content was transferred into this new TS 37.579-5 as by definition 36 series specifications cover LTE only aspects and multi-RAT aspects need to be covered in 37 series specifications. The only changes, compared to TS 36.579-5 v17.7.0, are: - "over LTE" has been removed from the TS title as beginning with Rel-17 the present document needs to cover also NR/5GC. - text in Scope clause has been updated, not to limit the RAT to LTE. - every instance of 36.579 has been replaced by 37.579, except in the Change history table of TS 36.579-5. - every instance of 36.579-3 has been either deleted or voided. - the CR history table of TS 36.579-5 was kept for easier reference of all changes included in TS 37.579-5, but a new Change history table was added for TS 37.579-5. - the 3GPP TS-TR template version 1.18.1 has been used.	0.0.1
2024-11	RAN5#105	R5-246420	-	-	-	36.579-5 CR#0123 has been implemented on top of 37.579-5-001 to produce 37.579-5-010.	0.1.0
2024-12	RAN#106	RP-242531	-	-	-	For one-step approval at RAN#106	1.0.0
2024-12	RAN#106	RP-243251	-	-	-	The associated TTCN-3 code has been added as an attachment	1.0.1
2024-12	RAN#106	-	-	-	-	raised to v17.0.0 with no change	17.0.0
2024-12	RAN#106	-	-	-	-	raised to v18.0.0 with no change	18.0.0
2025-03	RAN#107	R5-250360	0007	-	F	Routine maintenance for TS 37.579-5	18.1.0
2025-03	RAN#107	R5-251407	0006	1	F	Addition of initial Test Model for support of MCX Client over NR/5GC	18.1.0
2025-03	RAN#107	R5s250034	0001	-	F	Rel-18 Jun'24 baseline upgrade for MCX Test Suites	18.1.0
2025-03	RAN#107	R5s250035	0002	-	F	Correction of f_MCX_JSON_CreateTokens	18.1.0
2025-03	RAN#107	R5s250036	0003	-	F	Rel-18 Sep'24 baseline corrections for MCX Test Suites	18.1.0
2025-03	RAN#107	R5s250037	0004	-	B	Addition of MCPTT test case 5.1 to the MCX ATS with MCX-EUTRA model	18.1.0
2025-03	RAN#107	R5s250038	0005	-	F	Integrity protection added to NOTIFY for xcap-diff	18.1.0
2025-03	RAN#107	R5s250078	0009	-	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.579-5 (prose), Annex A	18.1.0
2025-06	RAN#108	R5-251937	0011	-	F	NR/5GC enhancement of MCX IPCAN Test Model	18.2.0
2025-06	RAN#108	R5-252226	0012	-	F	Routine maintenance for TS 37.579-5	18.2.0
2025-06	RAN#108	R5-252297	0013	-	F	Addition of Functional Alias PIXIT for MCVideo	18.2.0
2025-06	RAN#108	R5s250133	0008	1	B	Addition of MCPTT test case 6.1.1.16 to the MCX ATS with MCX-EUTRA model	18.2.0
2025-06	RAN#108	R5s250157	0010	-	F	Correction to function f_MCX_Registration_PDN_List_Match	18.2.0
2025-06	RAN#108	R5s250219	0015	-	F	Update of TTCN type definitions, Annex E	18.2.0
2025-09	RAN#109	R5-254019	0017	-	F	Routine maintenance for TS 37.579-5	18.3.0
2025-09	RAN#109	R5s250357	0018	-	F	Update of TTCN type definitions, Annex E	18.3.0
2025-12	RAN#110	R5-255799	0022	-	F	Routine maintenance for TS 37.579-5	18.4.0
2025-12	RAN#110	R5-256721	0023	1	F	Addition of MCData notification channel details	18.4.0
2025-12	RAN#110	R5s250406	0016	1	B	Addition of MCPTT test case 5.1 to the MCX ATS with MCX-NR5GC model	18.4.0
2025-12	RAN#110	R5s250429	0019	1	B	Addition of MCPTT test case 6.1.1.16 to the MCX ATS with MCX-NR5GC model	18.4.0
2025-12	RAN#110	R5s250505	0024	1	F	Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.579-5 (prose), Annex A	18.4.0

History

Version	Date	Status
V18.0.0	February 2025	Publication
V18.1.0	April 2025	Publication
V18.2.0	July 2025	Publication
V18.3.0	October 2025	Publication
V18.4.0	February 2026	Publication